MBG 1201 Introduction to Modern Biology

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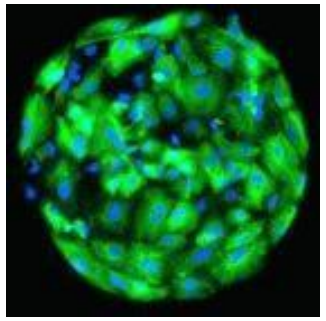
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10 December 2024

Outline

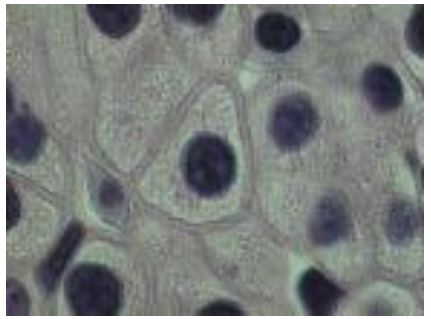
- Transcription: Initiation, elongation, termination
- Modification of mRNA
- Translation: Initiation, elongation, termination
- Point Mutations: Substitutions, insertions, deletions

What makes cells from the same individual look different?

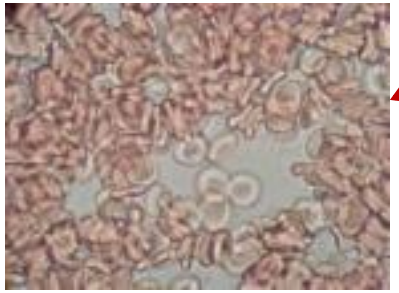


Stem Cells

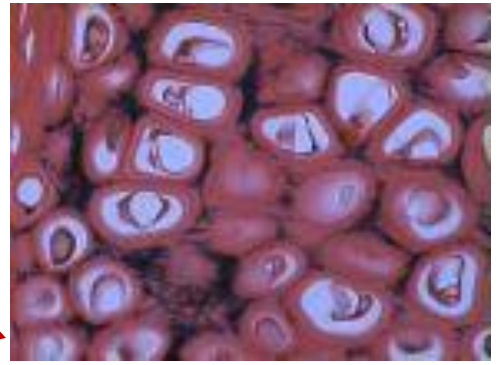




Liver Cells



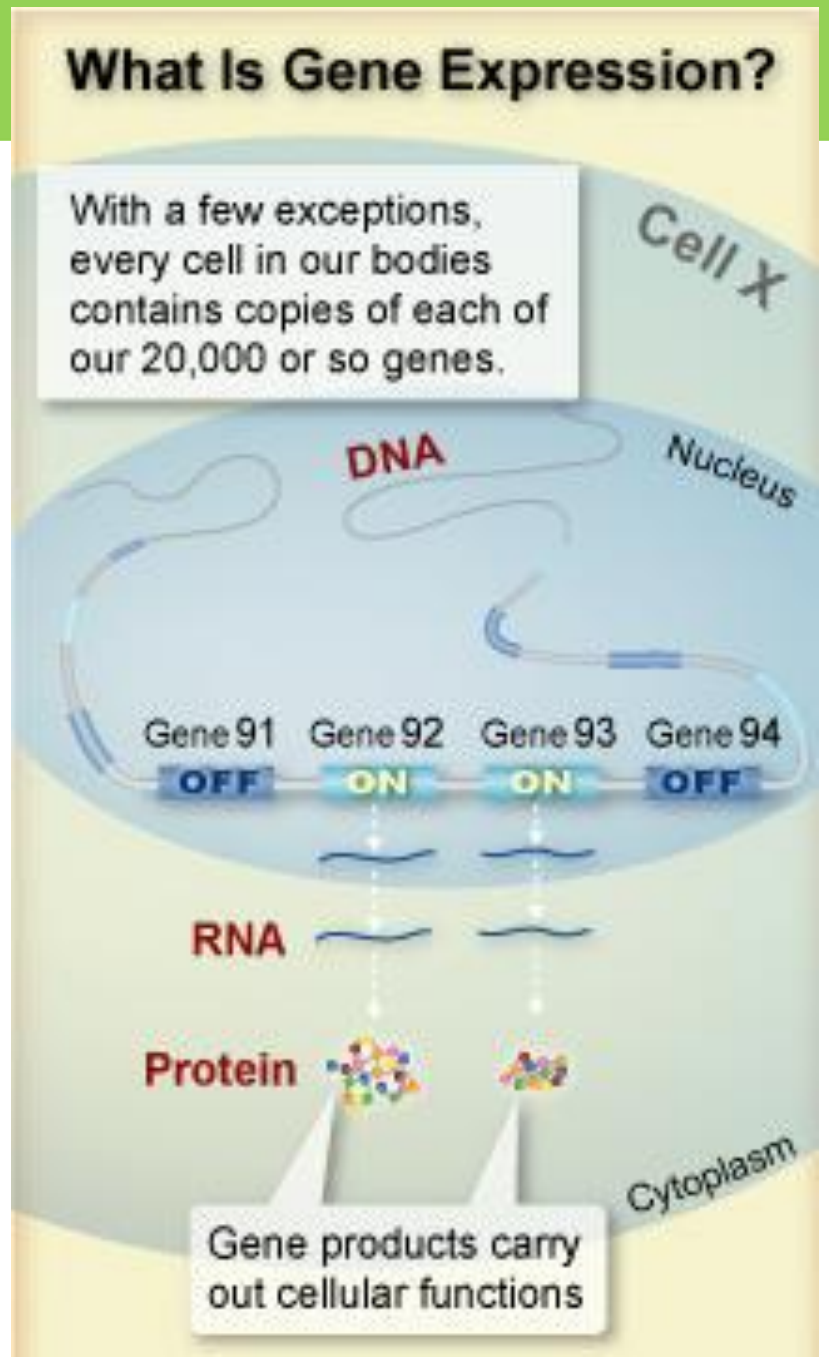
Red Blood Cells



Cartilage Cells

DNA sequence in each cell is the same, but different cell types have different "GENE EXPRESSION PATTERNS"

How do your cells "know" what kind of cell they are?
How do your cells "know" when to make a particular protein? When to stop making it?
How does the environment affect your cells?



- When a gene is "on" and its protein or RNA product is being made, scientists say that the gene is being EXPRESSED.
- The on and off states of all of a cell's genes is known as a GENE EXPRESSION PROFILE.
- Each cell type has a unique gene expression profile.

Insulin	DNA	Protein
Muscle Cell	✓	X
Pancreatic Cell	✓	✓

The first step of Gene Expression: Transcription

Transcription: the DNA directed synthesis of RNA.

Messenger RNA (mRNA: the carrier of information from DNA to the ribosomes (protein synthesis machinery). It is transcribed from the template strand of a gene.

RNA polymerase catalyzes transcription by separating the DNA strands and joining RNA nucleotides complementary to the DNA template strand.

Bacteria has only one type of RNA polymerase to transcribe all types of RNAs. Eukaryotes has at least 3 types of RNA polymerases. The one that synthesizes **mRNA** is called **RNA polymerase II**.

The first step of Gene Expression: Transcription

TABLE 6–1 Principal Types of RNAs Produced in Cells

Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids
snRNAs	Small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	Small nucleolar RNAs, help to process and chemically modify rRNAs
miRNAs	MicroRNAs, regulate gene expression by blocking translation of specific mRNAs and cause their degradation
siRNAs	Small interfering RNAs, turn off gene expression by directing the degradation of selective mRNAs and the establishment of compact chromatin structures
piRNAs	Piwi-interacting RNAs, bind to piwi proteins and protect the germ line from transposable elements
lncRNAs	Long noncoding RNAs, many of which serve as scaffolds; they regulate diverse cell processes, including X-chromosome inactivation

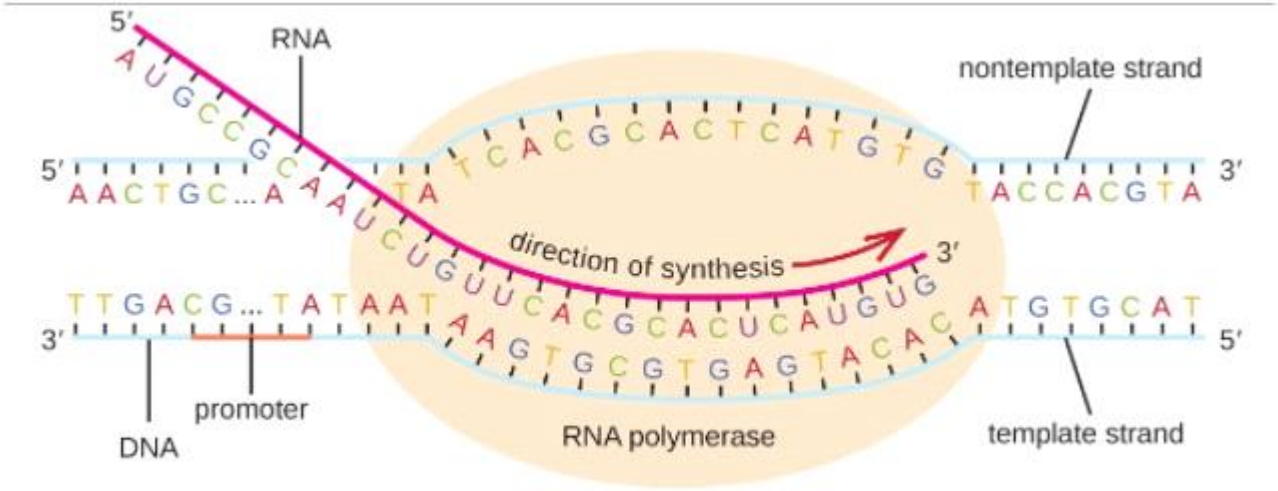
You only need to know the first 3.

The first step of Gene Expression: Transcription

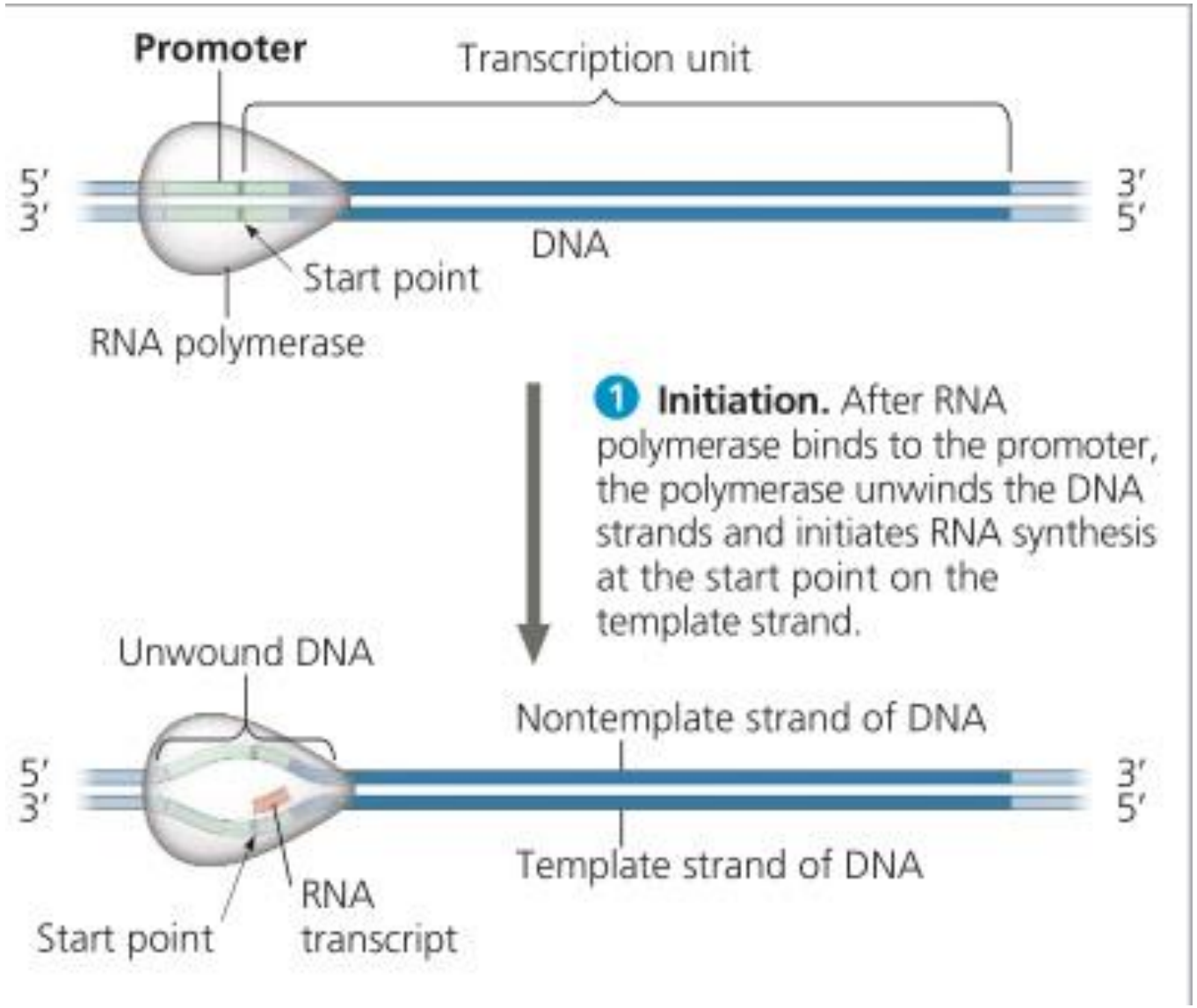
RNA polymerases **are able to start a chain from scratch**; they don't need to add the first nucleotide onto a pre-existing primer.

RNA polymerases can add nucleotides to the 3' end, thus assemble a polynucleotide only in its **5' to 3'** direction.

RNA synthesis follows the same base-pairing rules as DNA, except **uracil** substitutes for **thymine**.



Transcription: initiation



Start Point

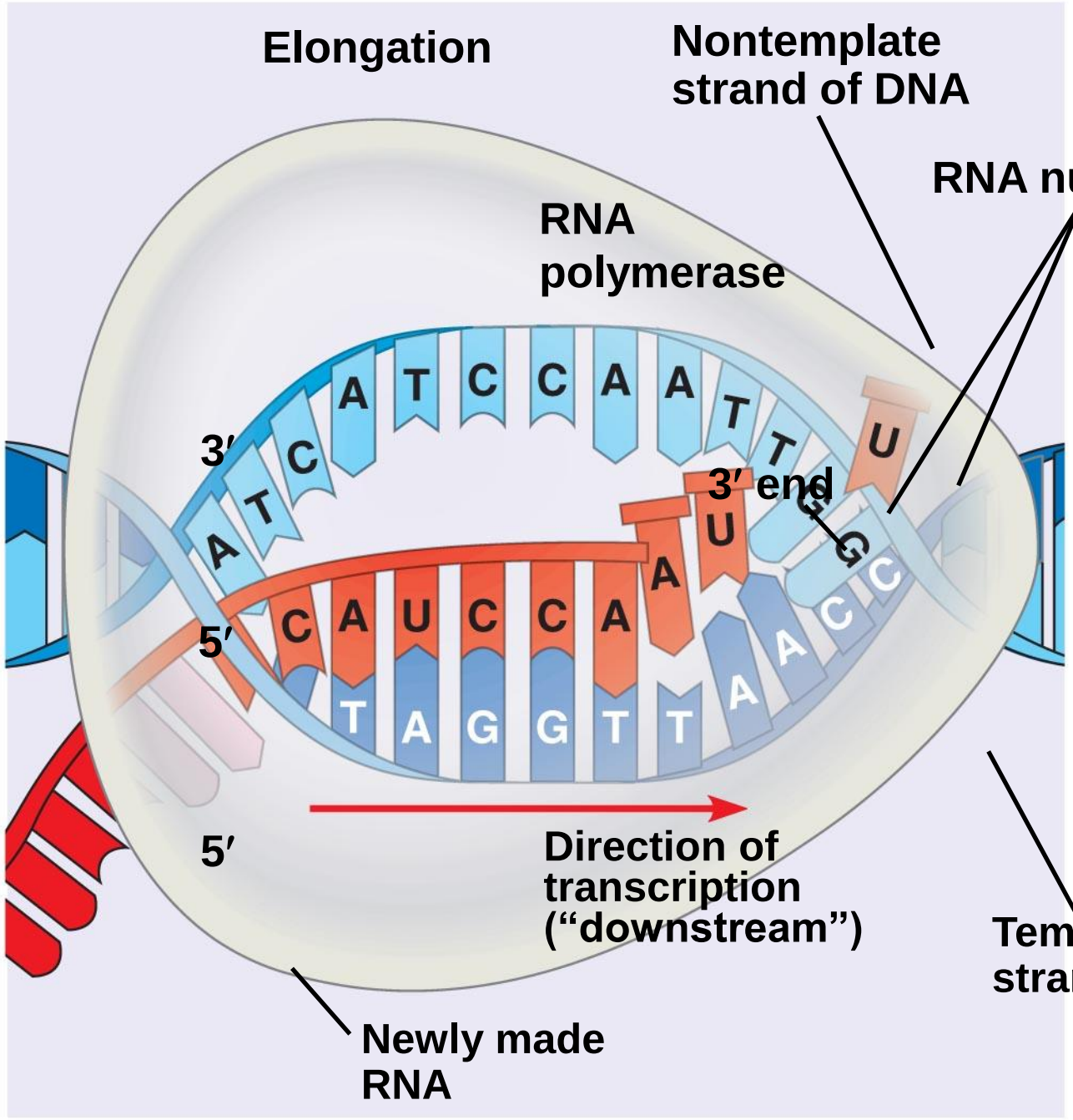
The promoter : The DNA sequence where RNA polymerase attaches and initiates transcription.

The terminator: in bacteria, the sequence that signals the end of transcription.

End Point in Bacteria

Transcription unit: The stretch of DNA that is transcribed.

Transcribed Part

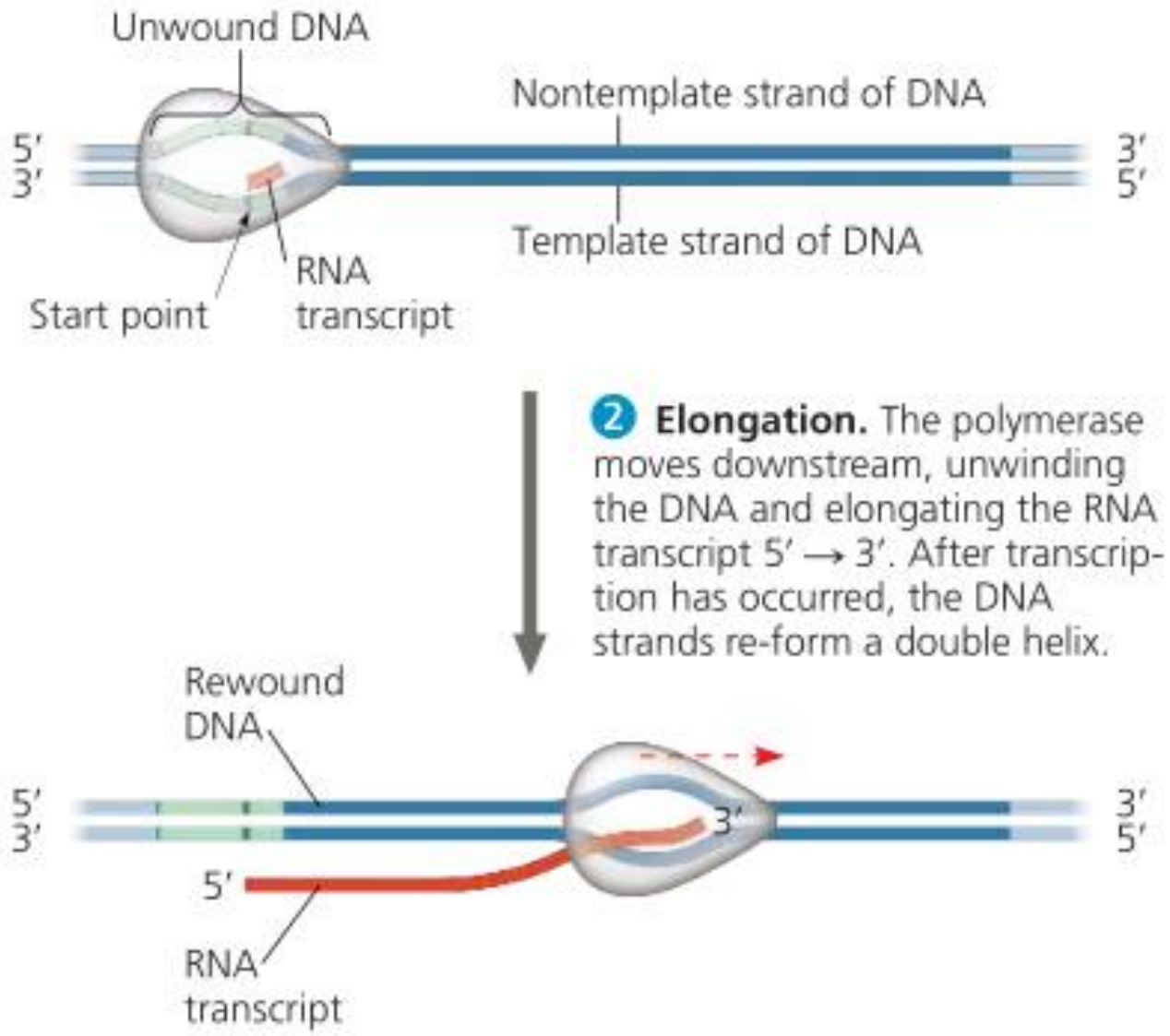


The direction of transcription is referred to as **downstream**. The other direction is referred to as **upstream**.

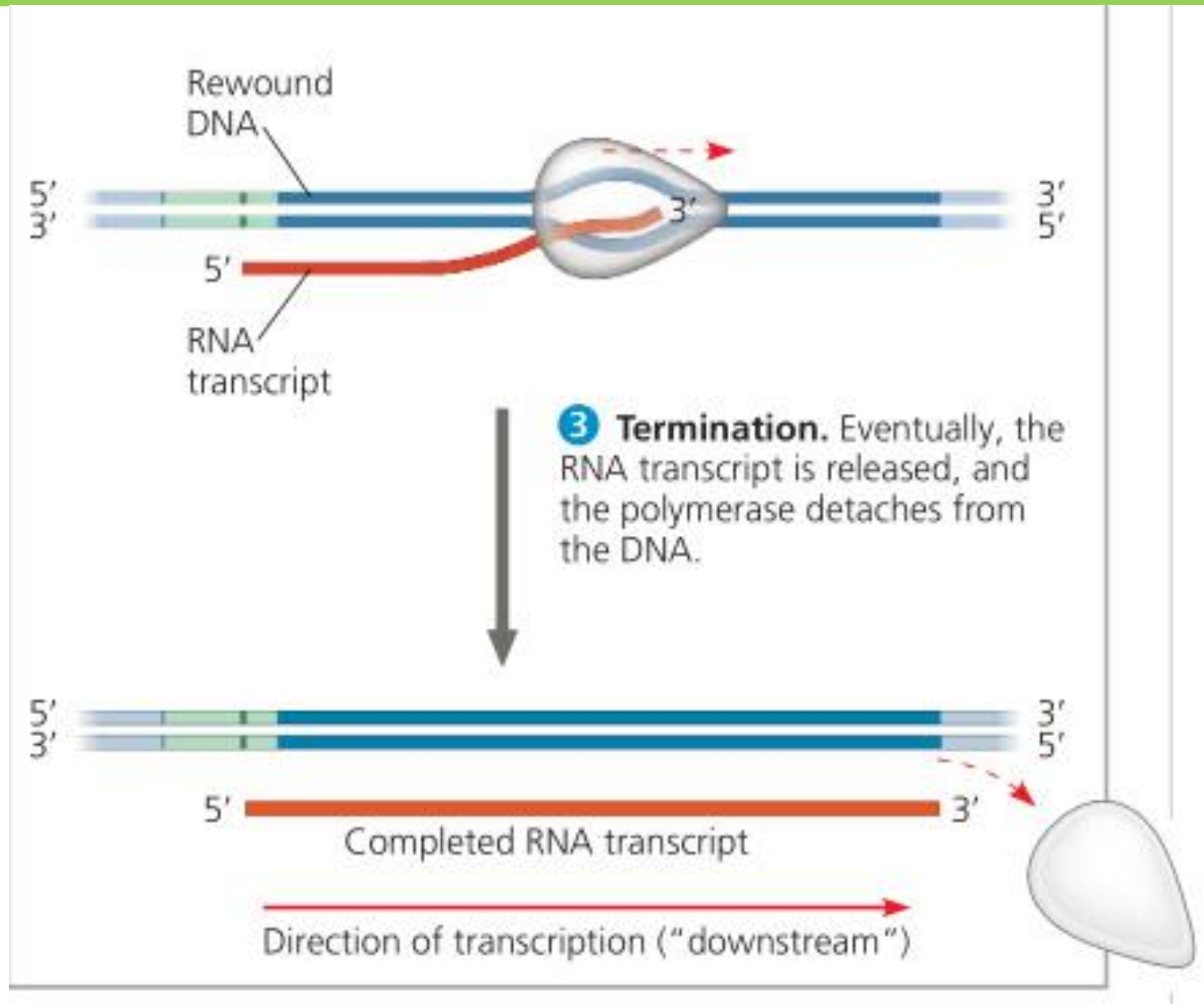
For example, the promoter is upstream to the terminator.

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Transcription: elongation



Transcription: termination



This general depiction of transcription applies to both bacteria and eukaryotes, but the details of termination differ.

Also, in a bacterium, the RNA transcript is immediately usable as mRNA; in a eukaryote, the RNA transcript must first undergo processing.

The initiation of transcription at the promoter

Promoters signal the initiation of RNA synthesis.

The presence of a promoter sequence determines which strand of the DNA helix is the template.

The starting point for the transcription of a gene is located within the promoter.

The promoter also includes **a binding site for RNA polymerase** several dozen nucleotides from the start point.

The initiation of transcription at the promoter

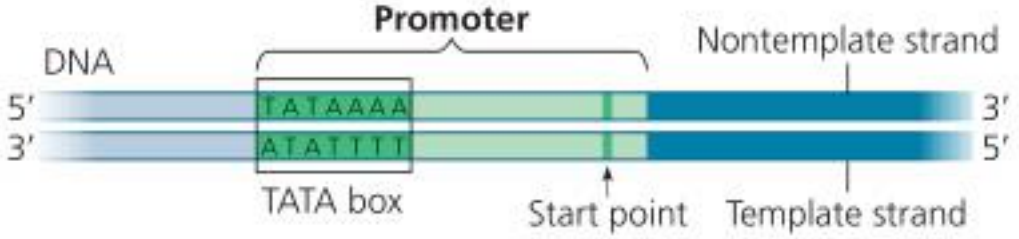
In bacteria, part of the RNA polymerase itself specifically recognizes and binds to the promoter. In eukaryotes, a collection of proteins called transcription factors mediate the binding of RNA polymerase and the initiation of transcription.

Only after certain transcription factors are attached to the promoter does RNA polymerase II bind to it and start transcription.

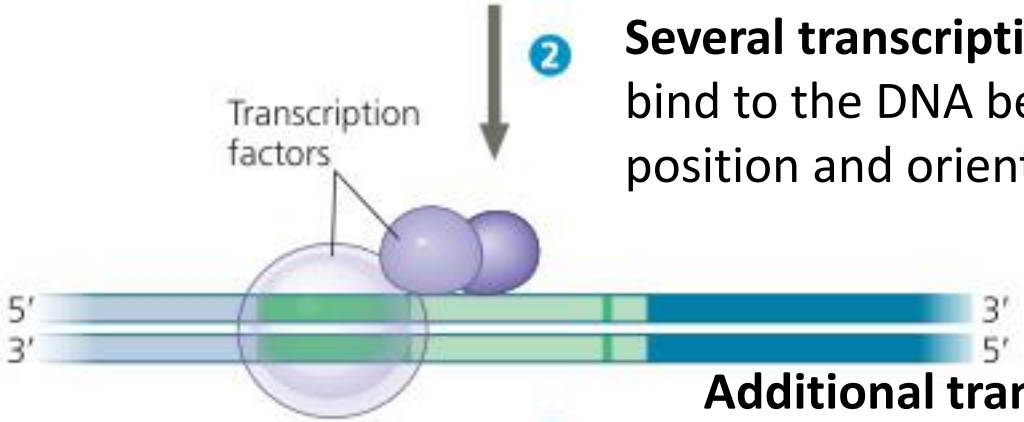
The completed assembly of transcription factors and RNA polymerase II bound to a promoter is called a **transcription initiation complex.**

A promoter called a **TATA box** is crucial in forming the initiation complex in eukaryotes.

The initiation of transcription at a eukaryotic promoter

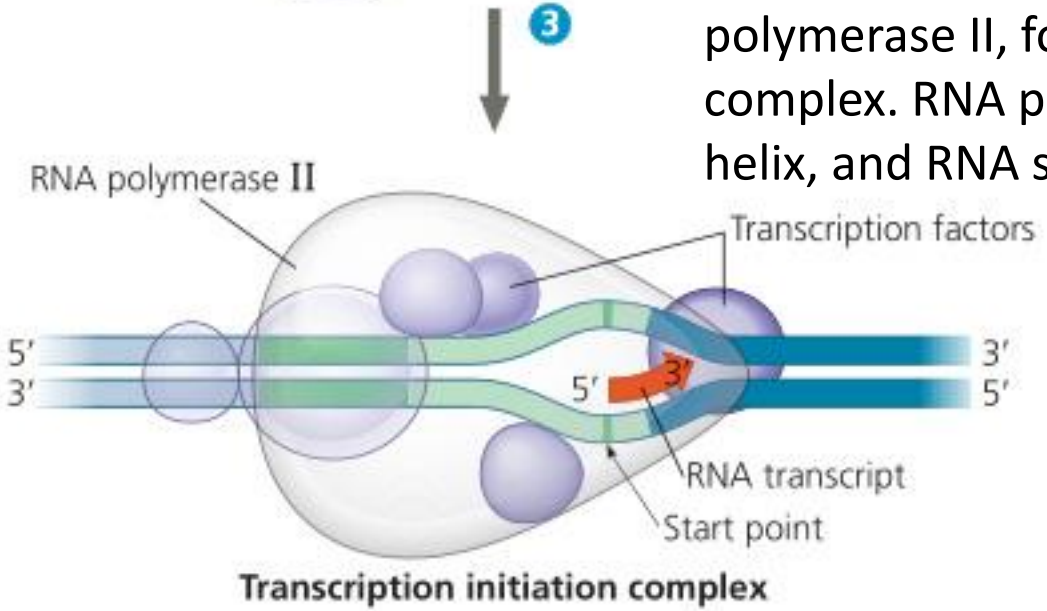


1 **A eukaryotic promoter** commonly includes a TATA box (a nucleotide sequence containing TATA) about 25 nucleotides upstream from the transcriptional start point.



Several transcription factors, one recognizing the TATA box, must bind to the DNA before RNA polymerase II can bind in the correct position and orientation, as shown in step 3.

Additional transcription factors (purple) bind to the DNA along with RNA polymerase II, forming the transcription initiation complex. RNA polymerase II then unwinds the DNA double helix, and RNA synthesis begins at the start point on the template strand.

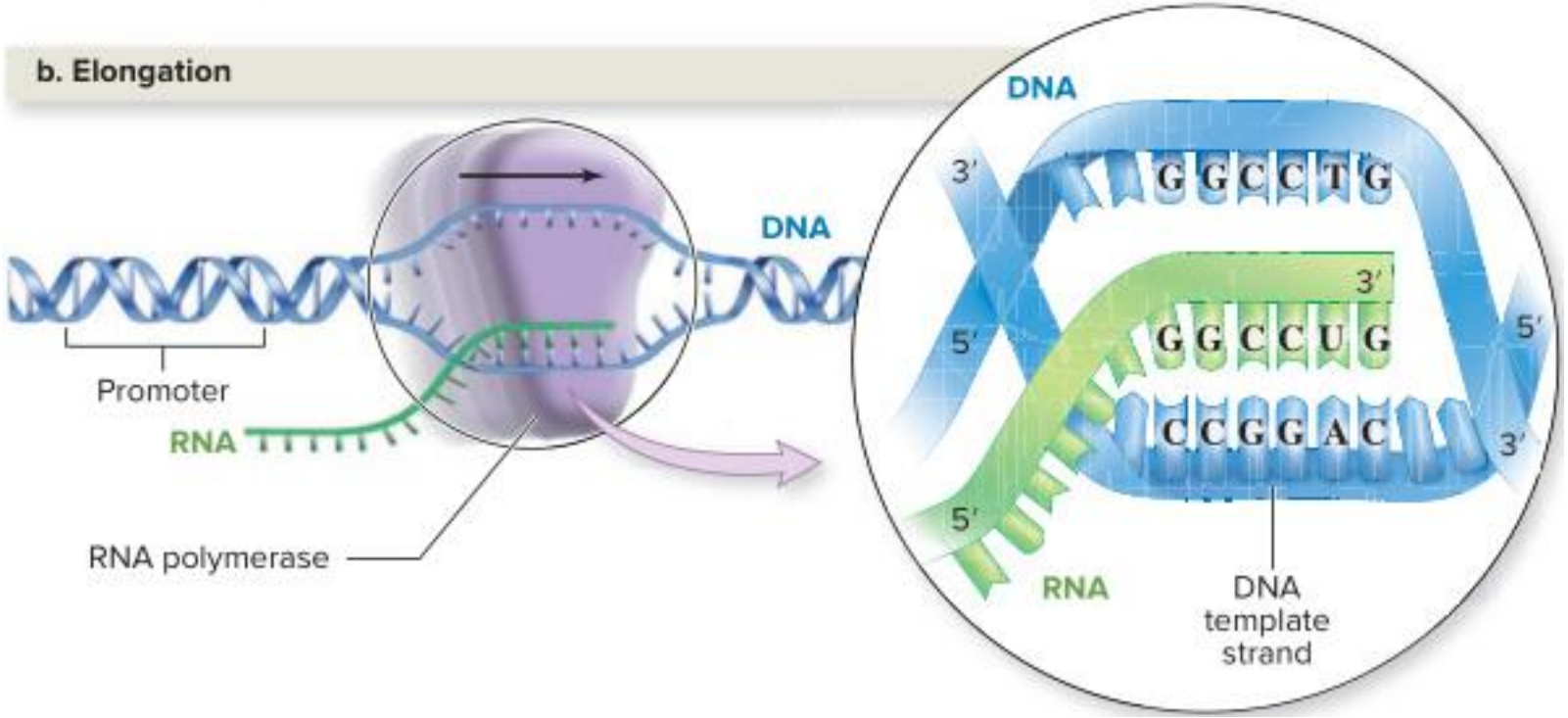


Elongation of the RNA Strand

As RNA polymerase moves along the DNA, it untwists the double helix, exposing about 10–20 DNA nucleotides at a time for pairing with RNA nucleotides

The enzyme adds nucleotides to the 3′ end of the growing RNA molecule as it moves along the double helix.

As transcription proceeds forward, the newly synthesized RNA molecule peels away from its DNA template, and the DNA double helix reforms.



Transcription progresses at a rate of 40 nucleotides per second in eukaryotes

A gene can be transcribed simultaneously by several RNA polymerases

Termination of Transcription

The mechanisms of termination are different in bacteria and eukaryotes

In bacteria, the polymerase stops transcription at the end of the terminator. The transcribed terminator (an RNA sequence) functions as the termination signal, causing the polymerase to detach from the DNA and release the transcript. The mRNA can be translated without further modification.

In eukaryotes, the pre-mRNA is cleaved from the growing RNA chain while RNA polymerase II continues to transcribe the DNA.; the polymerase eventually falls off the DNA.

- The RNA polymerase transcribes a DNA sequence called the polyadenylation signal sequence, (AAUAAA) in the pre-mRNA.
- At a point about 10 to 35 nucleotides past this sequence, the pre-mRNA is cut from the enzyme.
- The completed single-stranded RNA transcript is released and the RNA polymerase will detach from the DNA strand.

Alteration of mRNA Ends

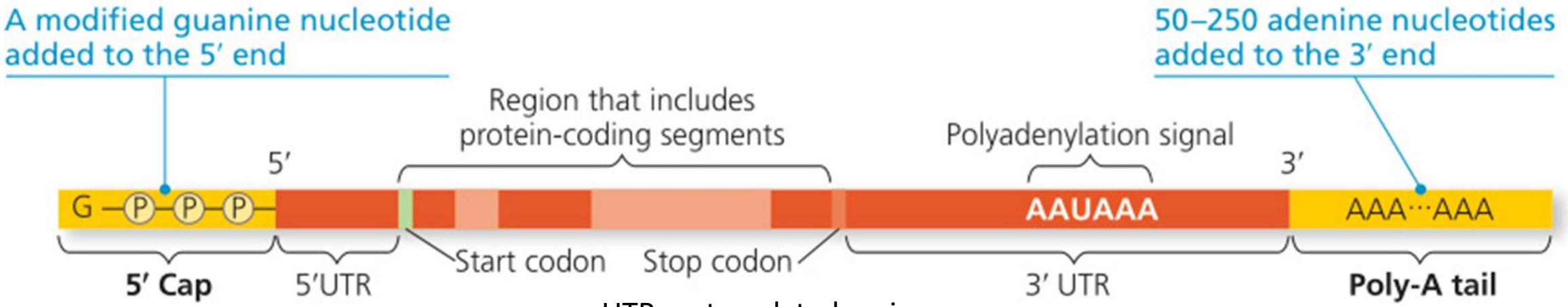
Before transfer into the cytoplasm mRNA is modified:
Each end of a pre-mRNA molecule is modified in a particular way:

The 5’ end which is synthesized first, receives a 5’ cap, a modified form of a guanine (G) nucleotide added onto the 5’ end after transcription of the first 20–40 nucleotides

The 3’ end gets a poly-A tail.

These modifications share several functions:

1. They seem to facilitate the export of mature mRNA from the nucleus
2. They protect mRNA from hydrolytic enzymes
3. They help ribosomes attach to the 5’ end



UTR: untranslated region

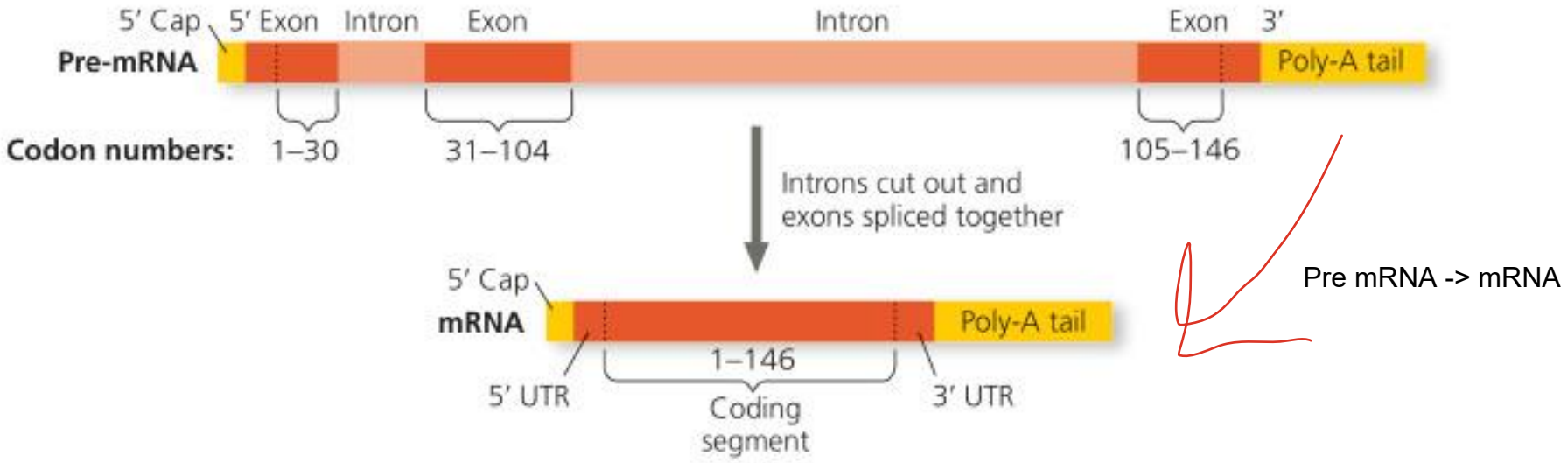
Split Genes and RNA Splicing

Most eukaryotic genes and their RNA transcripts have long noncoding stretches of nucleotides that lie between coding regions

These noncoding regions are called intervening sequences, or **introns**

The other regions are called **exons** because they are eventually expressed, usually translated into amino acid sequences

RNA splicing removes introns and joins exons, creating an mRNA molecule with a continuous coding sequence.

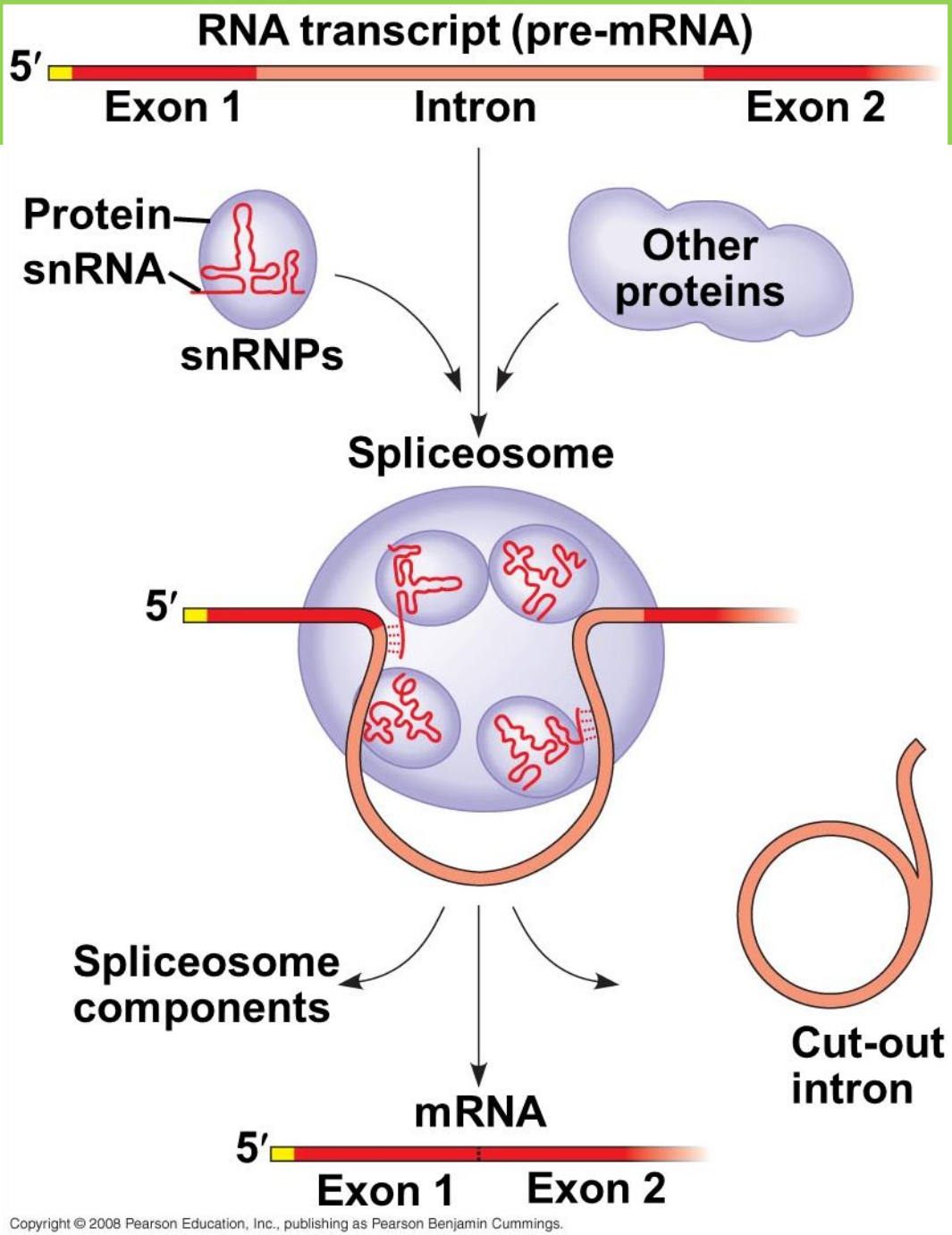


Split Genes and RNA Splicing

RNA splicing is carried out by **spliceosomes**

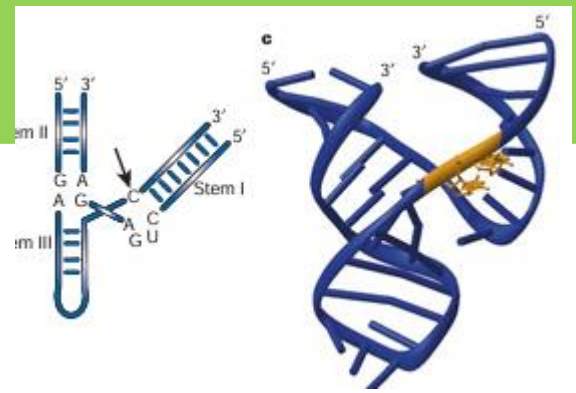
Spliceosomes consist of a variety of proteins and several small nuclear ribonucleoproteins (snRNPs) that recognize the splice sites

RNAs in the spliceosome not only participate in spliceosome assembly and splice site recognition, but also catalyze the splicing reaction; like proteins, RNAs can act as catalysts.



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Ribozymes



Three properties of RNA enable it to function as an enzyme:

1. It can form a 3-D structure because of its ability to base pair with itself. (A specific structure is essential to the catalytic function of ribozymes, just as it is for enzymatic proteins.)

2. Some bases in RNA contain functional groups that can participate in catalysis

3. RNA may hydrogen-bond with other nucleic acid molecules. This adds specificity to its catalytic activity. For example, complementary base pairing between the RNA of the spliceosome and the RNA of a primary RNA transcript precisely locates the region where the ribozyme catalyzes splicing.

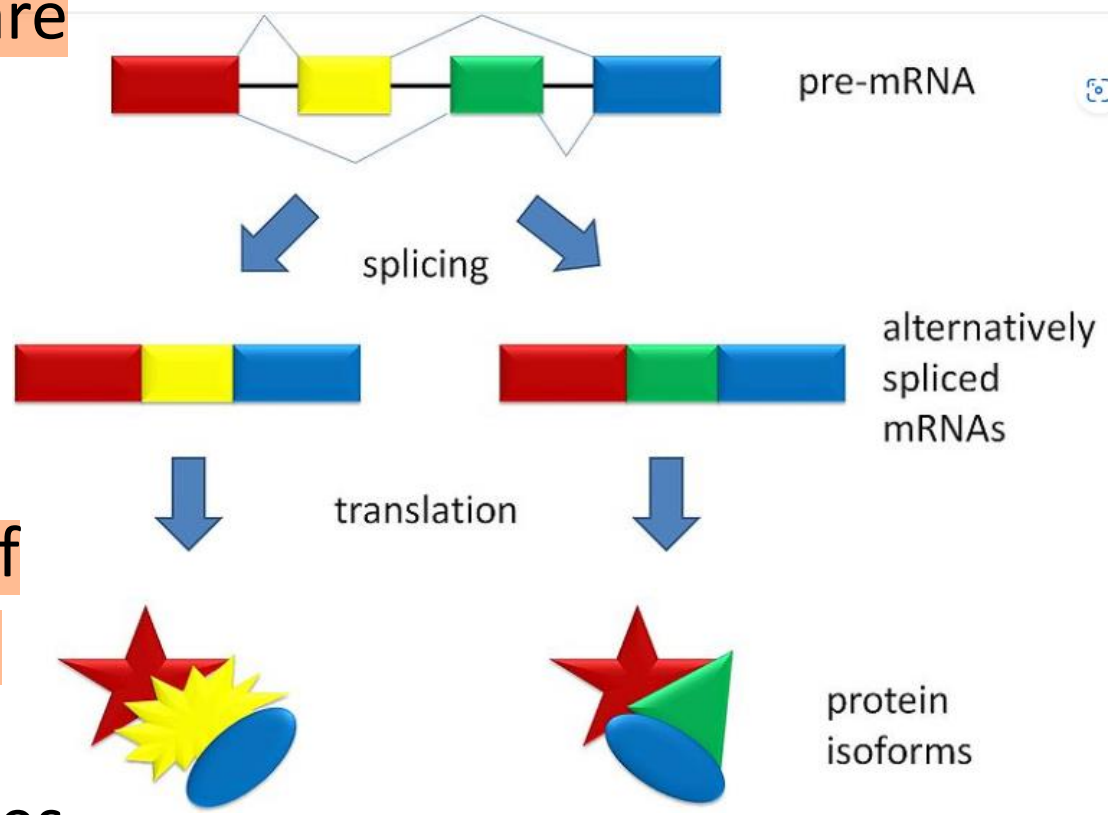
The Functional and Evolutionary Importance of Introns

Some genes can encode more than one kind of polypeptide, depending on which segments are treated as exons during RNA splicing.

Such variations are called **alternative RNA splicing**.

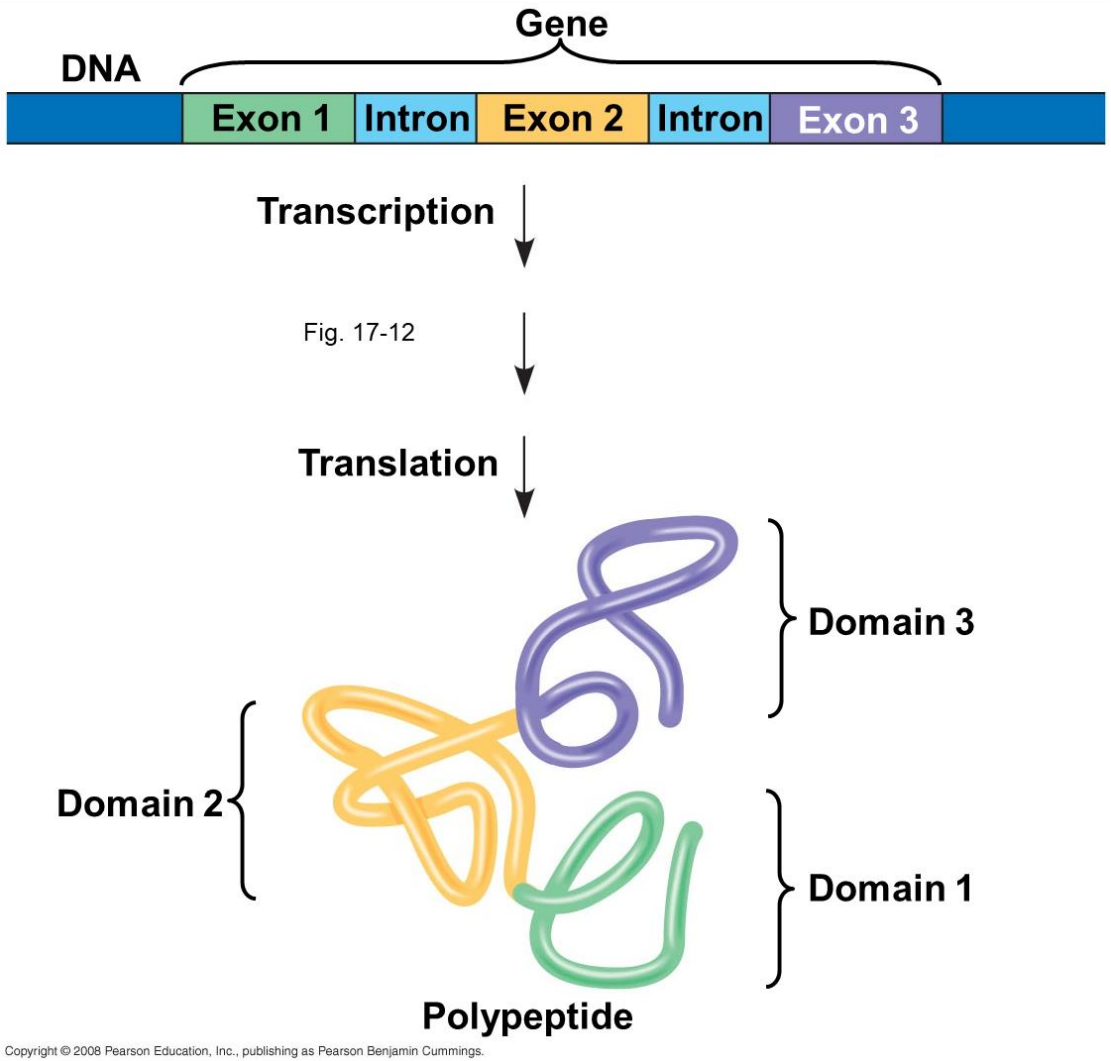
Because of alternative splicing, the number of different proteins an organism can produce is much greater than its number of genes.

Humans have about the same number of genes as the roundworm but produce a larger variety of proteins due to alternative splicing.



The Functional and Evolutionary Importance of Introns

Proteins often have a modular architecture consisting of discrete regions called **domains**
In many cases, different exons code for the different domains in a protein
The presence of introns in a gene may facilitate the evolution of new and potentially beneficial proteins as a result of a process known as exon shuffling.
Introns increase the probability of crossing over between the exons
of alleles of a gene—simply by providing more terrain for crossovers without interrupting coding sequences.



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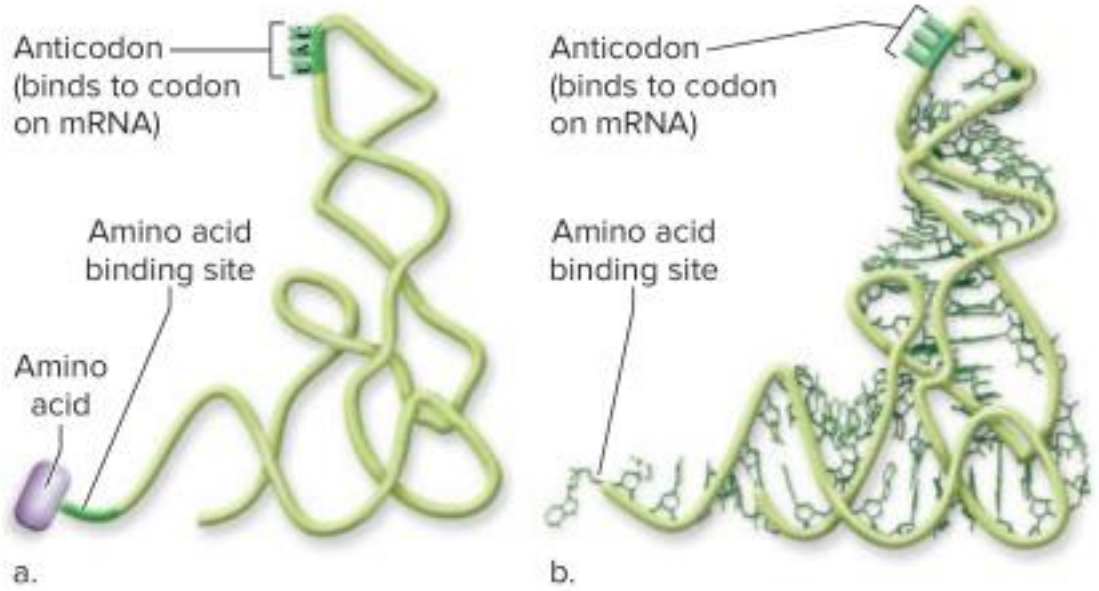
Translation: the RNA-directed synthesis of a polypeptide

A cell translates an mRNA message into protein with the help of transfer RNA (tRNA).

Molecules of tRNA are not identical:
Each carries a specific amino acid on one end.
Each has an anticodon on the other end; the anticodon base-pairs with a complementary codon on mRNA.

Example: If the codon on mRNA is UUU, a tRNA with an AAA anticodon and carrying phenylalanine amino acid will bind to it.

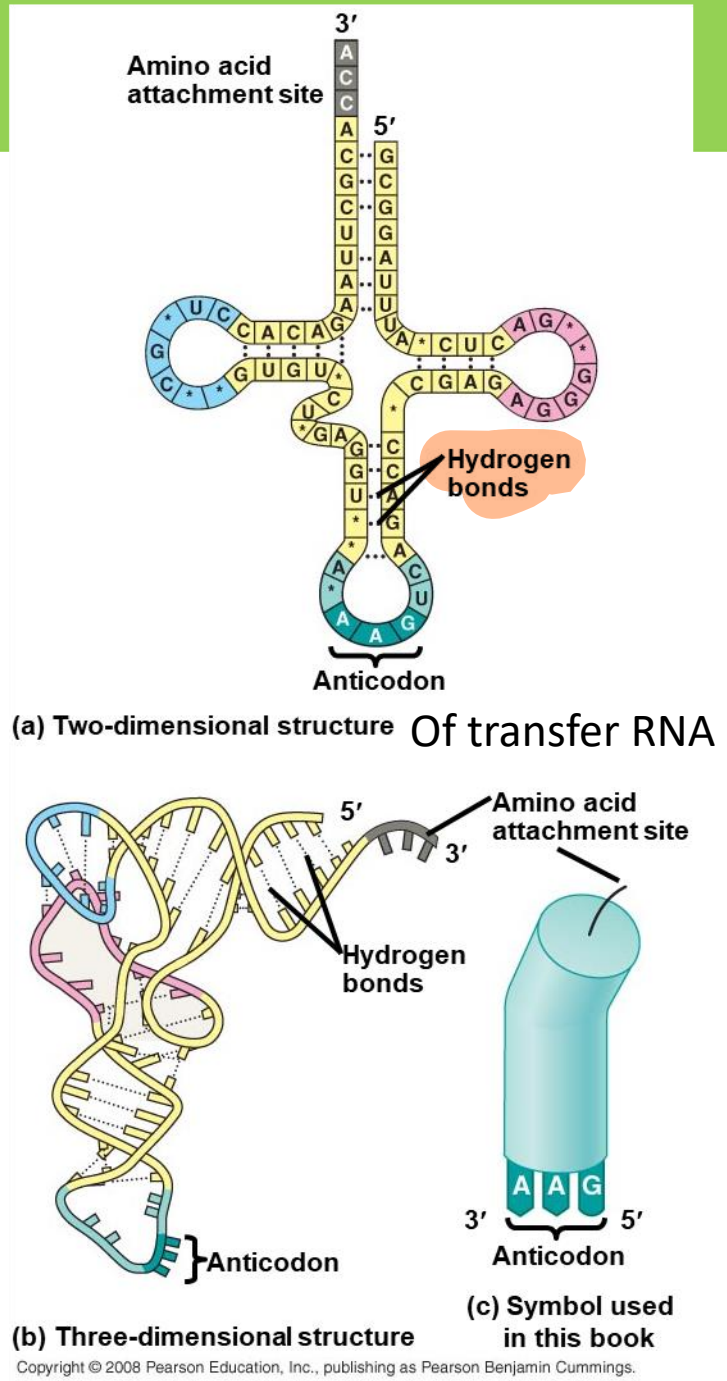
- Codon by codon, tRNAs deposit amino acids in the prescribed order, and the ribosome joins them into a polypeptide chain.



The Structure and Function of Transfer RNA

- tRNA is transcribed in the nucleus from a DNA template and must enter the cytoplasm, where the ribosomes are located.
- Each tRNA is used repeatedly, picking up its designated amino acid in the cytosol, depositing the amino acid at the ribosome, and returning to the cytosol to pick up another copy of that amino acid.
- A tRNA molecule consists of a single RNA strand that is only about 80 nucleotides long.
- Flattened into one plane to reveal its base pairing, a tRNA molecule looks like a cloverleaf.

Because of hydrogen bonds, tRNA actually twists and folds into a three-dimensional molecule which is roughly L-shaped.

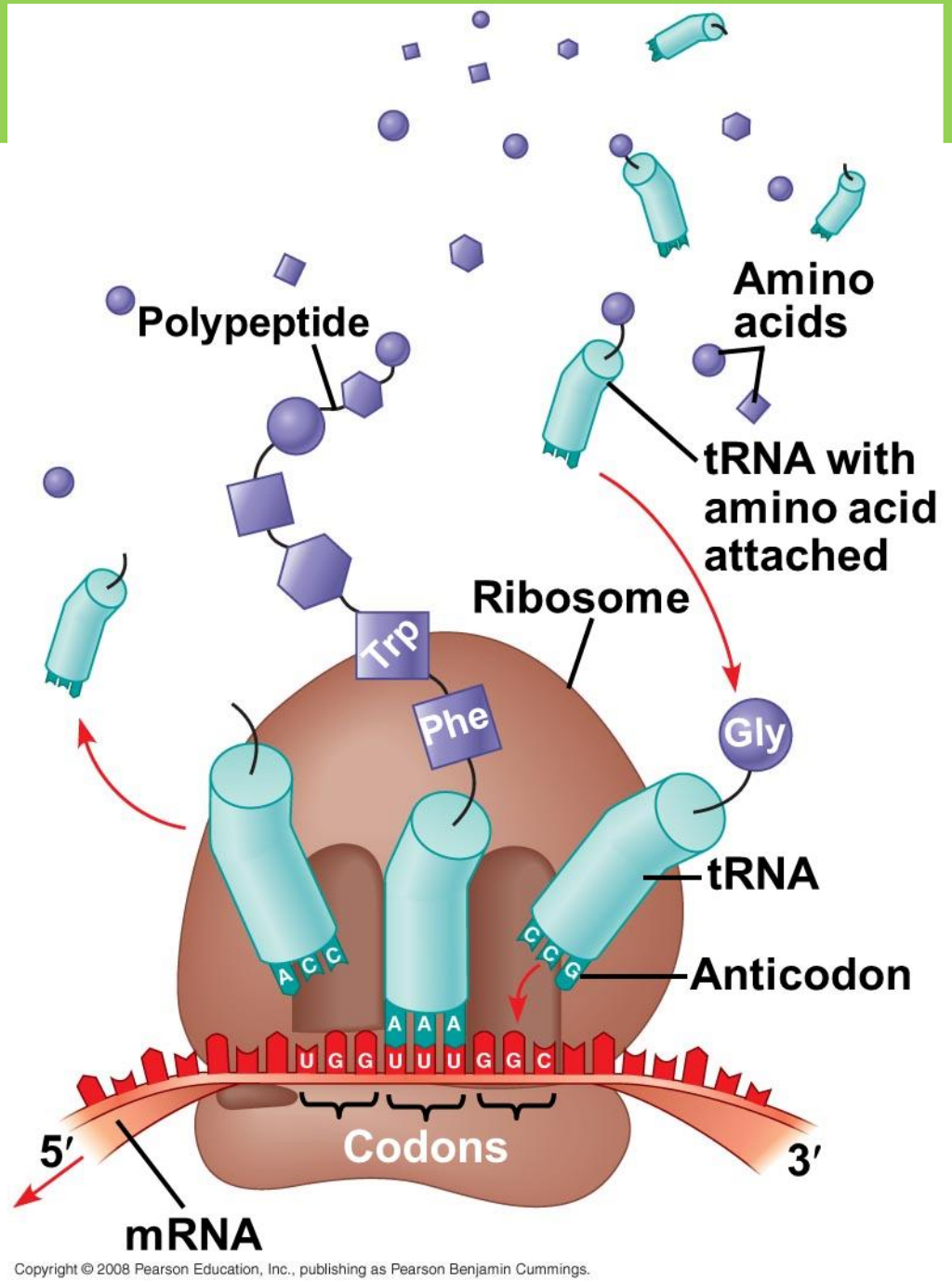


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Translation

As a molecule of mRNA moves through a ribosome, codons are translated into amino acids one at a time.

- tRNA have a specific anticodon of 3 nucleotide complementary sequence to the mRNA on one end and the corresponding amino acid at the other end.
- When H bonding between the anticodon and mRNA occurs, the ribosome adds the amino acid to the growing polypeptide chain.



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Translation

Accurate translation requires two steps:

- a correct match between a tRNA and an amino acid, done by the enzyme **aminoacyl-tRNA synthetase.**

- a correct match between the tRNA anticodon and an mRNA codon

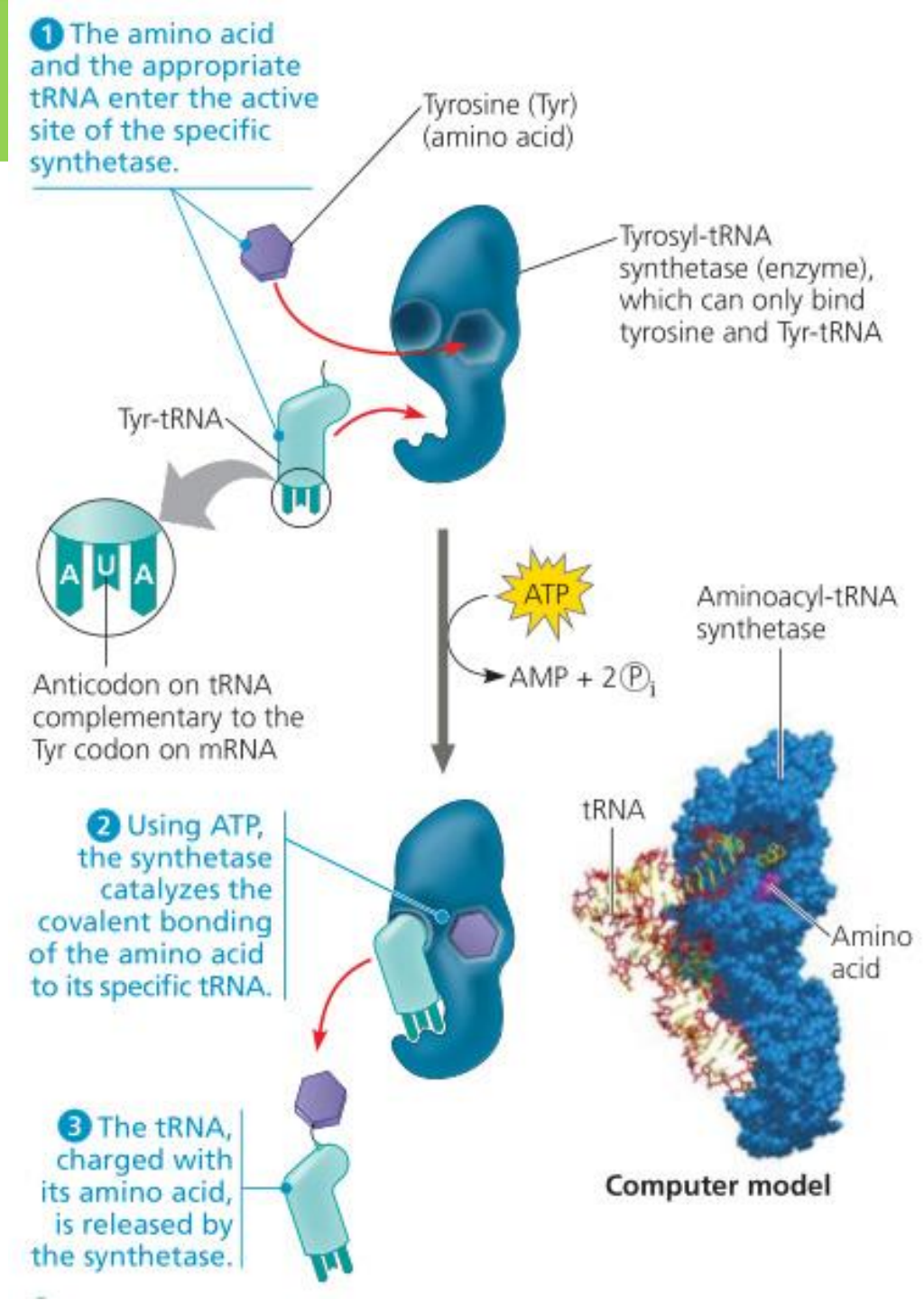
Translation

Aminoacyl-tRNA synthetases provide specificity in joining amino acids to their tRNAs.

There are 20 different synthetases, one for each amino acid. The active site of each type of aminoacyl-tRNA synthetase fits only a specific combination of amino acid and tRNA.

A synthetase joins a given amino acid to an appropriate tRNA covalently in a process driven by the hydrolysis of ATP.

The resulting aminoacyl tRNA, also called a charged tRNA, is released from the enzyme and is then available to deliver its amino acid to a growing polypeptide chain on a ribosome.



Wobble

If one tRNA type existed for each mRNA codon specifying an amino acid, there would be 61 tRNAs . In bacteria, however, there are only about 45 tRNAs, signifying that some tRNAs must be able to bind to more than one codon.

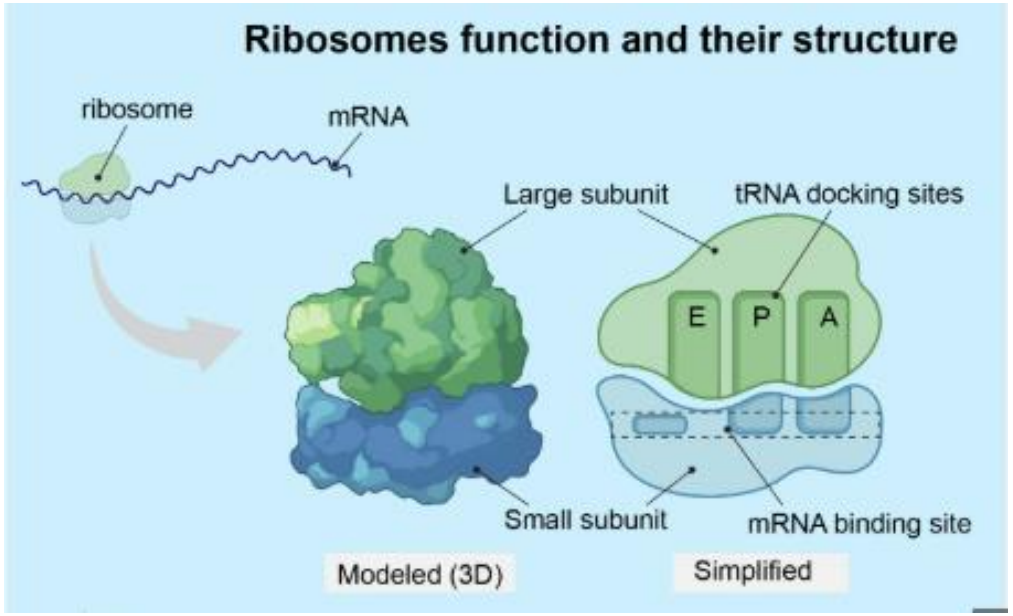
Flexible pairing at the third base of a codon is called **wobble** and allows some tRNAs to bind to more than one codon. Wobble explains why the synonymous codons for a given amino acid most often differ in their third nucleotide base. Accordingly, a tRNA with the anticodon 3'UCU5' can base-pair with either the mRNA codon 5'AGA3' or 5'AGG3', both of which code for arginine

Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
GCA	AGA	GAC	AAC	UGC	GAA	CAA	GGA	CAC	AUA	UUA	AAA	AUG	UUC	CCA	AGC	ACA	UGG	UAC	GUA	UAA
GCC	AGG	GAU	AAU	UGU	GAG	CAG	GGC	CAU	AUC	UUG	AAG		UUU	CCC	AGU	ACC		UAU	GUC	UAG
GCG	CGA						GGG		AUU	CUA				CCG	UCA	ACG			GUG	UGA
GCU	CGC						GGU			CUC				CCU	UCC	ACU			GUU	
	CGG									CUG					UCG					
	CGU									CUU					UCU					

The Structure and Function of Ribosomes

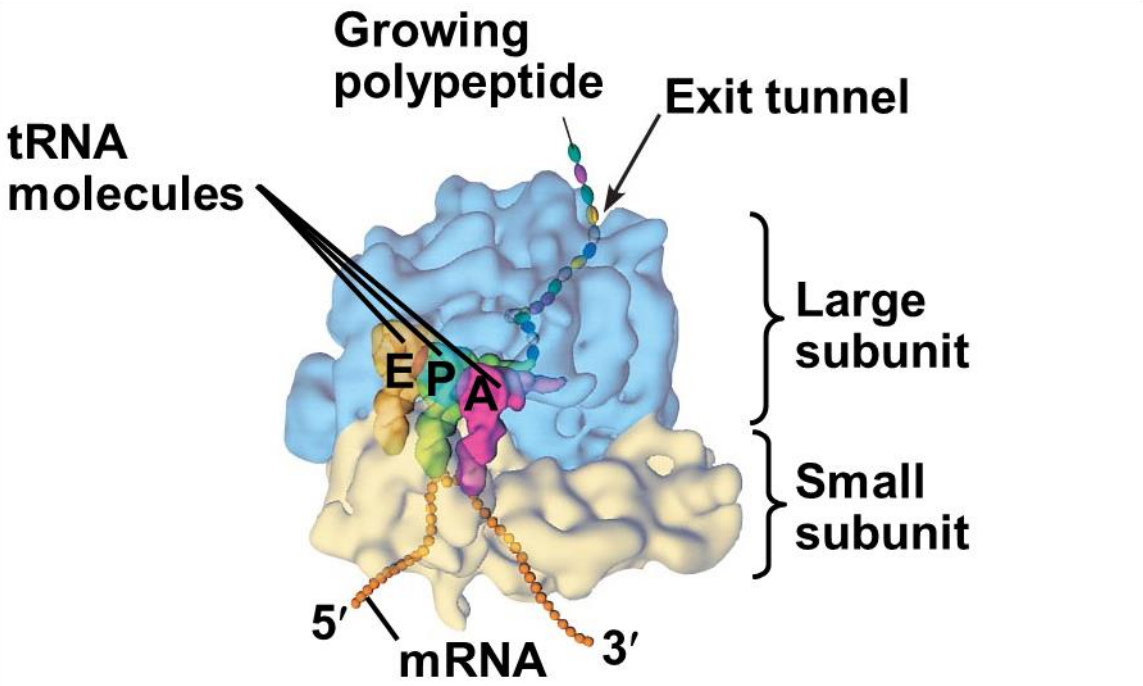
Ribosomes facilitate the specific coupling of tRNA anticodons with mRNA codons during protein synthesis.

A ribosome consists of a large subunit and a small subunit, each made up of proteins and one or more ribosomal RNAs (rRNAs).



In both bacteria and eukaryotes, a large and a small subunit join to form a functional ribosome only when attached to an mRNA molecule.

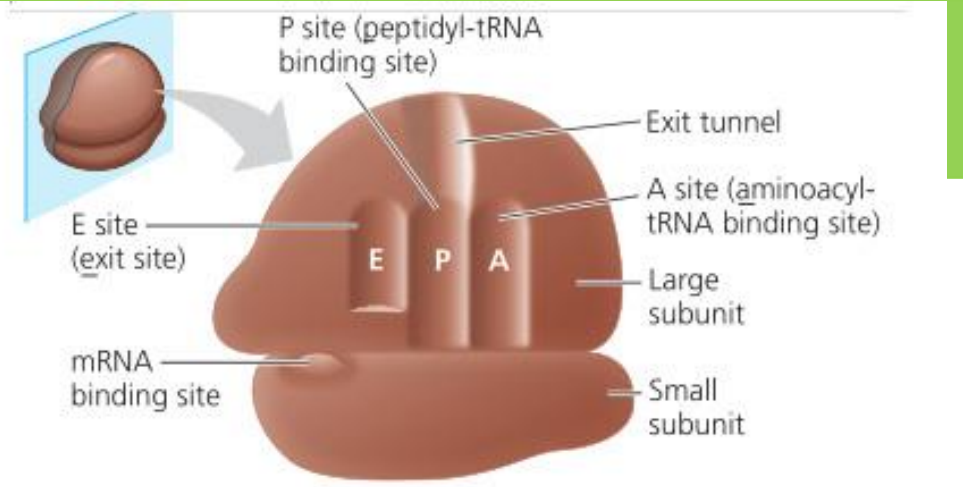
The Structure and Function of Ribosomes



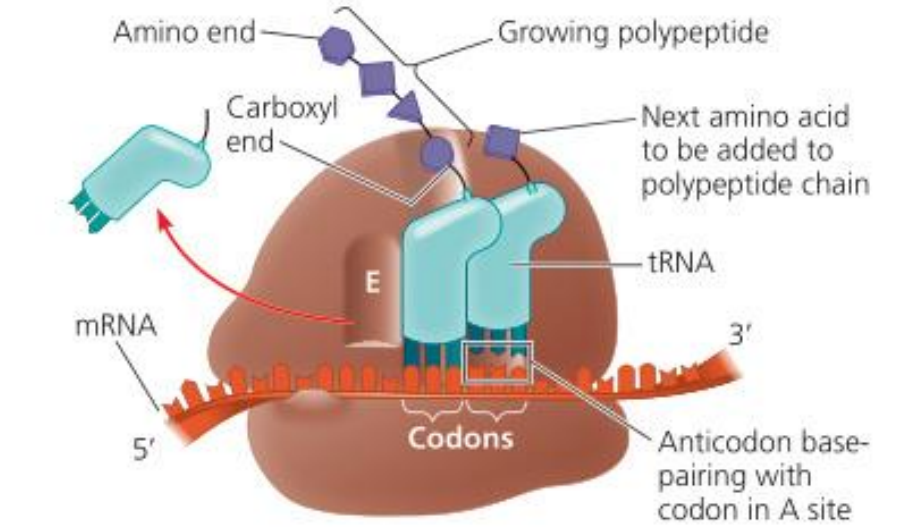
(a) Computer model of functioning ribosome

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The ribosome holds the tRNA and mRNA in close proximity and positions the new amino acid so that it can be added to the carboxyl end of the growing polypeptide. It then catalyzes the formation of the peptide bond. As the polypeptide becomes longer, it passes through an exit tunnel in the ribosome's large subunit. When the polypeptide is complete, it is released through the exit tunnel.



(b) Schematic model showing binding sites. A ribosome has an mRNA binding site and three tRNA binding sites, known as the A, P, and E sites. This schematic ribosome will appear in later diagrams.



(c) Schematic model with mRNA and tRNA. A tRNA fits into the A site when its anticodon base-pairs with an mRNA codon. The P site holds the tRNA attached to the growing polypeptide. The A site holds the tRNA carrying the next amino acid to be added to the polypeptide chain. Discharged tRNAs leave from the E site. The polypeptide grows at its carboxyl end.

The Structure and Function of Ribosomes

A ribosome has three binding sites for tRNA:

The P site holds the tRNA that carries the growing polypeptide chain

The A site holds the tRNA that carries the next amino acid to be added to the chain

The E site is the exit site, where discharged tRNAs leave the ribosome

Ribosomal RNA is the main constituent of the A and P sites and of the interface between the two subunits; it also acts as the catalyst of peptide bond formation. Thus, a ribosome could actually be considered one colossal ribozyme!

Building a Polypeptide

The three stages of translation:

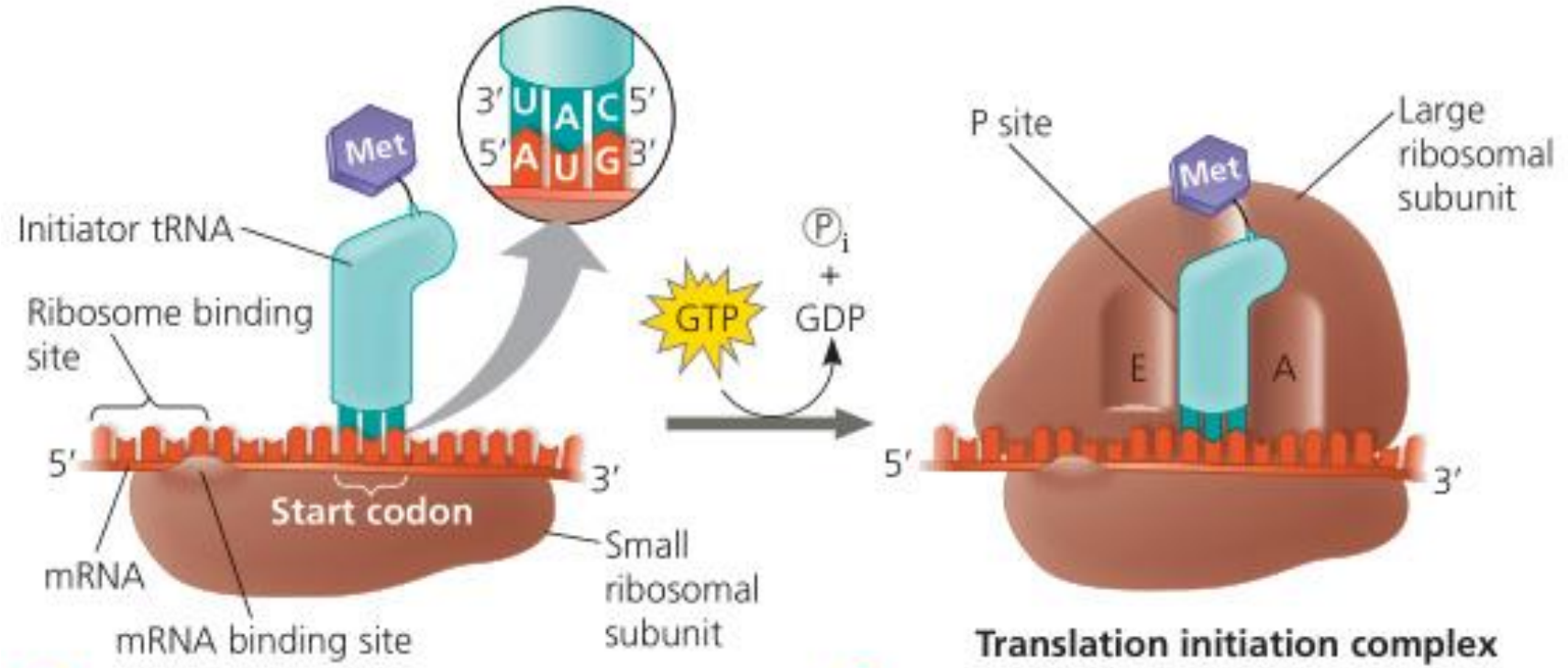
Initiation

Elongation

Termination

All three stages require protein “factors” that aid in the translation process. Some steps of initiation and elongation also require energy, provided by the **hydrolysis of guanosine triphosphate (GTP).**

Ribosome Association and Initiation of Translation



Translation initiation complex

1. A small ribosomal subunit binds to an mRNA. In a bacterial cell, the mRNA binding site on this subunit recognizes a specific nucleotide sequence on the mRNA just upstream of the start codon (see the Scientific Skills Exercise). An initiator tRNA, with the anticodon UAC, base-pairs with the start codon, AUG. This tRNA carries the amino acid methionine (Met).

2. The arrival of a large ribosomal subunit completes the initiation complex. Proteins called initiation factors (not shown) are required to bring all the translation components together. Hydrolysis of GTP provides the energy for the assembly. The initiator tRNA is in the P site; the A site is available to the tRNA bearing the next amino acid.

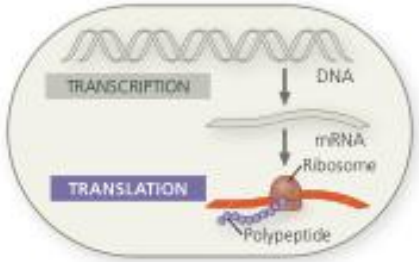
Ribosome Association and Initiation of Translation

First, The small subunit binds to (in eukaryotes the 5′ cap of) the mRNA and a special initiator tRNA, which carries methionine.

Then the small subunit moves downstream along the mRNA until it reaches the start codon AUG, which signals the start of translation.
This establishes the reading frame for the mRNA.

Proteins called initiation factors bring in the large subunit that completes the translation initiation complex.

The initiator tRNA, already associated with the complex, then hydrogen-bonds with the start codon.



Amino end of polypeptide

Carboxyl end of polypeptide

1 Codon recognition. The anticodon of an incoming aminoacyl tRNA base-pairs with the complementary mRNA codon in the A site. Hydrolysis of GTP increases the accuracy and efficiency of this step. Although not shown, many different aminoacyl tRNAs are present, but only the one with the appropriate anticodon will bind and allow the cycle to progress.

E

P site

A site

3′

mRNA 5′

Ribosome ready for next aminoacyl tRNA

GTP

GDP + P_i

2 Peptide bond formation. An rRNA molecule of the large ribosomal subunit catalyzes the formation of a peptide bond between the carboxyl end of the growing polypeptide in the P site and the amino group of the new amino acid in the A site. As shown in the next diagram, this step removes the polypeptide from the tRNA in the P site and attaches it to the amino acid on the tRNA in the A site.

GDP + P_i

GTP

3 Translocation. The ribosome translocates the tRNA in the A site to the P site. At the same time, the empty tRNA in the P site is moved to the E site, where it is released. The mRNA moves along with its bound tRNAs, bringing the next codon to be translated into the A site.

Elongation of the Polypeptide Chain

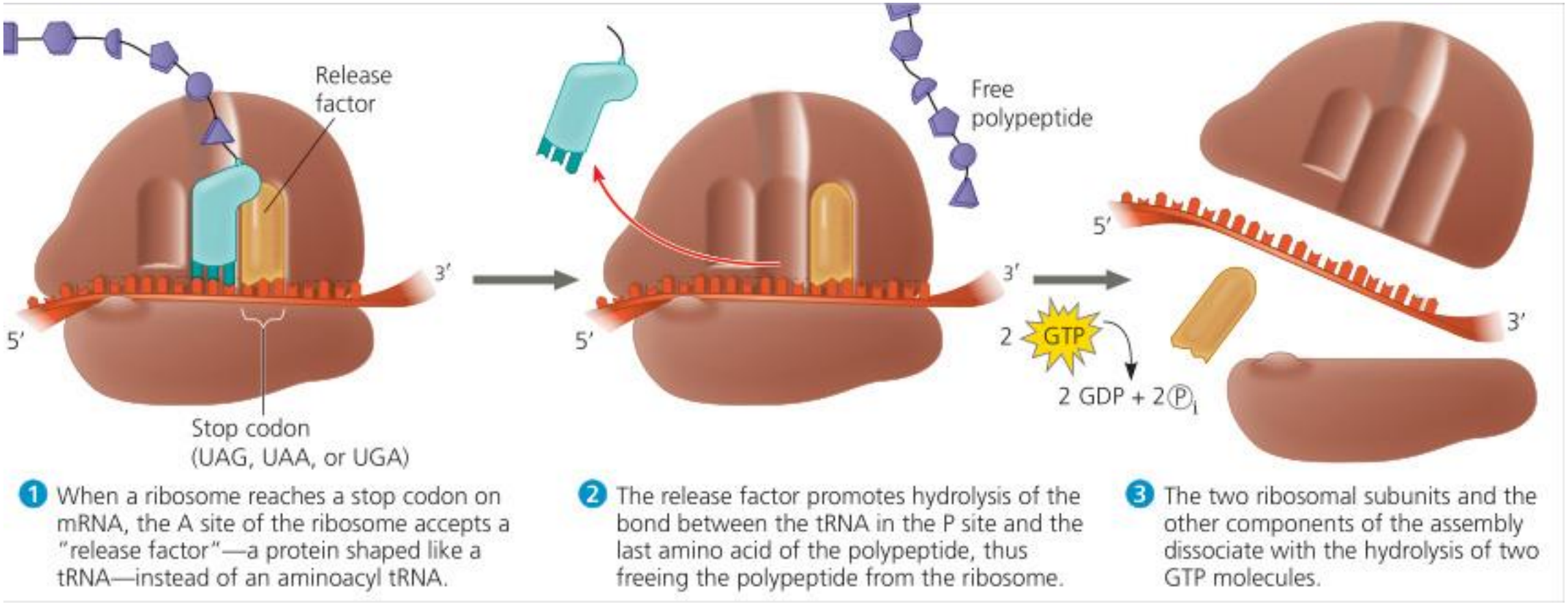
During the elongation stage, amino acids are added one by one to the preceding amino acid at the C-terminus of the growing chain.

Each addition involves several proteins called elongation factors and occurs in a three-step cycle: codon recognition, peptide bond formation, and translocation

Translation proceeds along the mRNA in a 5′ to 3′ direction.

The hydrolysis of GTP plays an important role in the elongation process. Codon recognition requires hydrolysis of
one molecule of GTP. One more GTP is hydrolyzed (broken up) to provide energy for the translocation step.

Termination of Translation



1. When a ribosome reaches a stop codon on mRNA, the A site of the ribosome accepts a "release factor"—a protein shaped like a tRNA—instead of an aminoacyl tRNA.
2. The release factor promotes hydrolysis of the bond between the tRNA in the P site and the last amino acid of the polypeptide, thus freeing the polypeptide from the ribosome.
3. The two ribosomal subunits and the other components of the assembly dissociate with the hydrolysis of two GTP molecules.

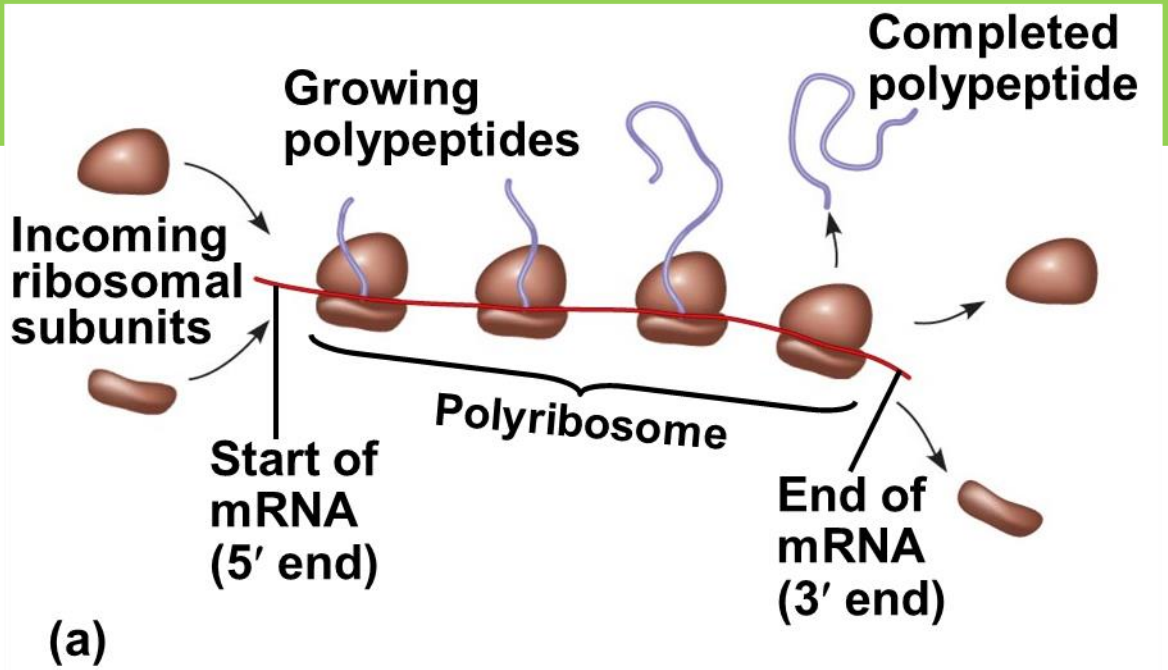
Termination of Translation

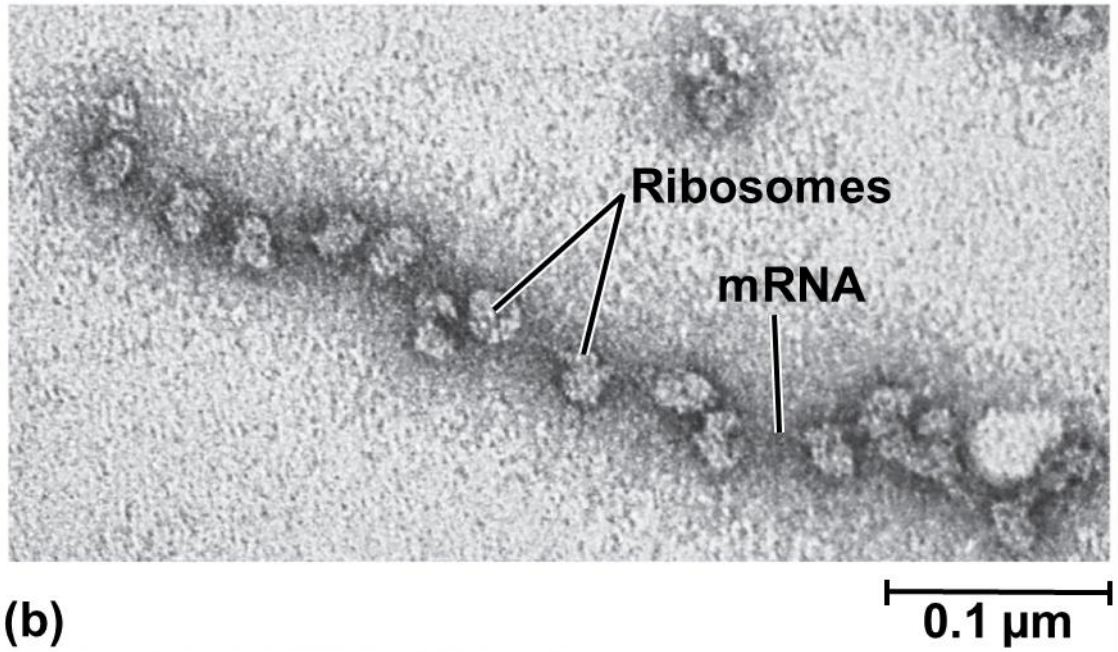
Termination occurs when a stop codon in the mRNA reaches the A site of the ribosome.
A release factor, a protein shaped like an aminoacyl tRNA, binds directly to the stop codon in the A site.
The release factor causes the addition of a water molecule instead of an amino acid to the peptide chain.
This reaction hydrolyzes (breaks) the bond between the completed polypeptide and the tRNA in the P site, releasing the polypeptide through the exit tunnel of the ribosome's large subunit.
The remainder of the translation assembly then comes apart in a multistep process, aided by other protein factors.
Breakdown of the translation assembly requires the hydrolysis of two more GTP molecules.

Polyribosomes

A number of ribosomes can translate a single mRNA simultaneously, forming a polyribosome (or polysome)
Polyribosomes enable a cell to make many copies of a polypeptide very quickly.

Another way both bacteria and eukaryotes increase the number of copies of a polypeptide is by transcribing multiple mRNAs from the same gene.

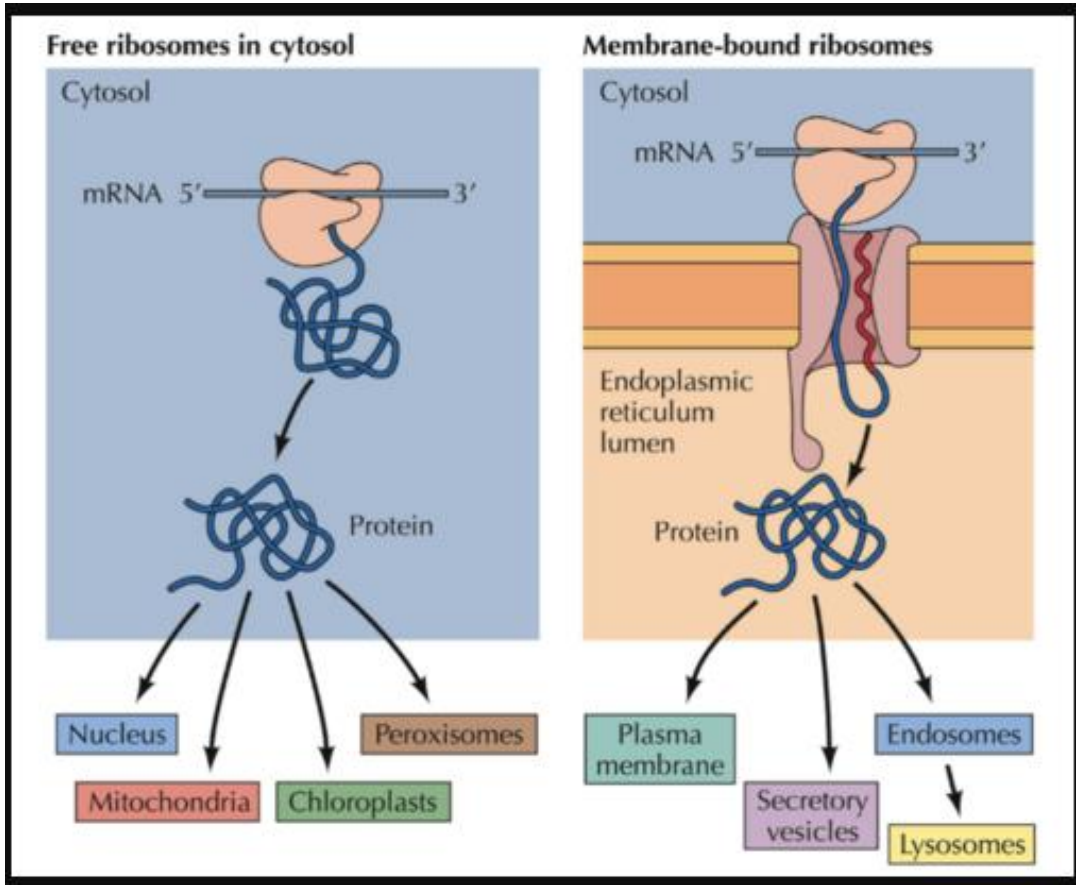




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Completing and Targeting the Functional Protein

Often translation is not sufficient to make a functional protein.
Polypeptide chains are modified after translation.
Completed proteins are targeted to specific sites in the cell.



Targeting Polypeptides to Specific Locations

Two populations of ribosomes are evident in cells: free ribsomes (in the cytosol) and bound ribosomes (attached to the ER)

Free ribosomes mostly synthesize proteins that function in the cytosol

Bound ribosomes make proteins of the endomembrane system and proteins that are secreted from the cell

Ribosomes are identical and can switch from free to bound

Targeting Polypeptides to Specific Locations

Polypeptide synthesis always begins in the cytosol.

– The polypeptide that is being synthesized will cause the ribosome to attach to the ER if it contains a **signal peptide sequence**.

Polypeptides of membrane proteins and proteins that are to be secreted contain signal peptide sequences.

– Signal peptides are around 20 amino acids long and are located near the amino end of the polypeptide chain.

• The signal peptide is recognized by a protein-RNA complex called a **signal-recognition complex.**

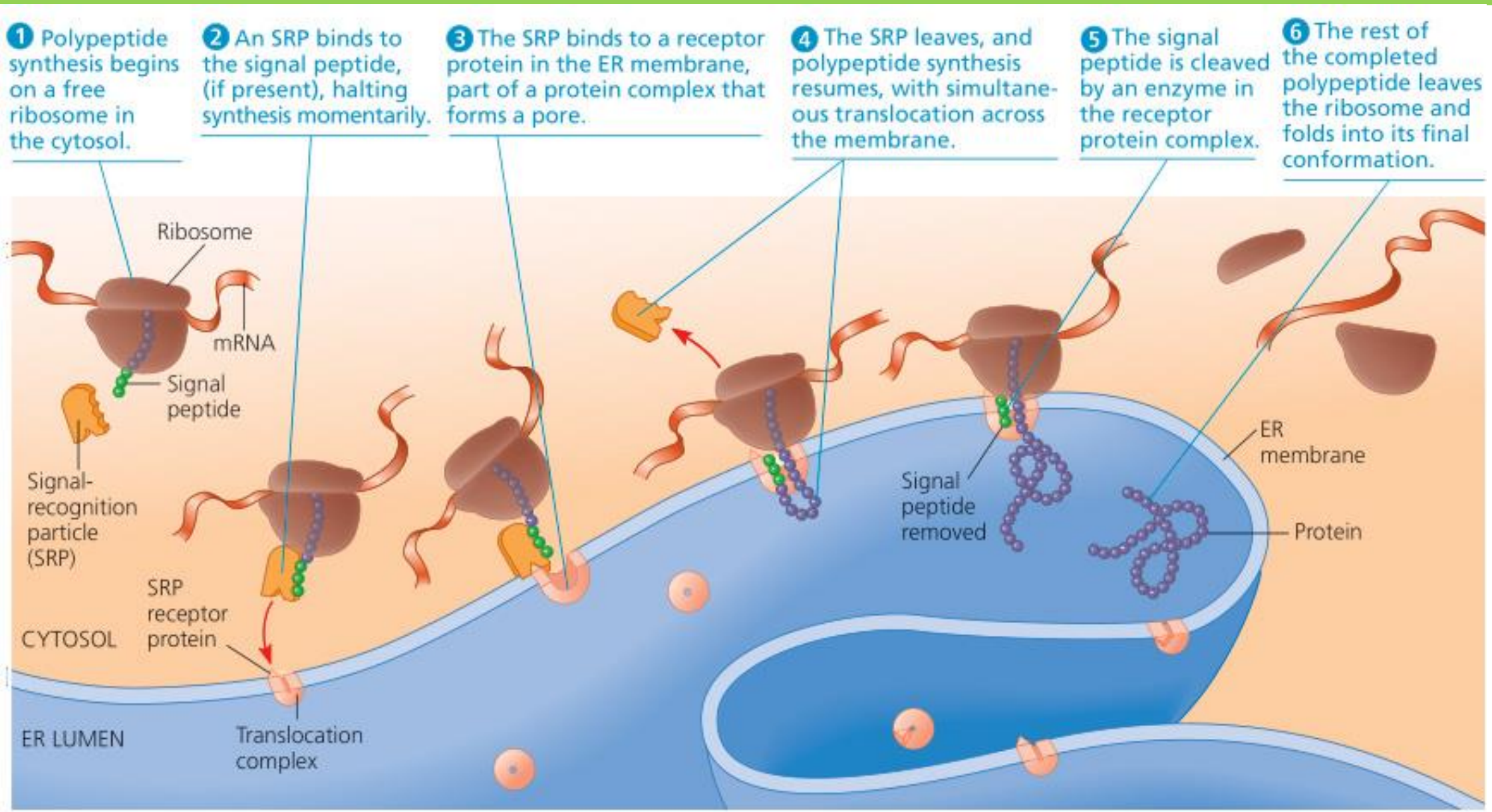
A signal-recognition particle (SRP) binds to the signal peptide

The SRP brings the signal peptide and its ribosome to the ER.

Targeting Polypeptides to Specific Locations

1. Polypeptide synthesis begins on a free ribosome in the cytosol.
2. An SRP binds to the signal peptide and temporarily stops polypeptide synthesis.
3. The SRP binds to a receptor protein in the ER membrane.
4. The SRP leaves and polypeptide synthesis resumes and translocation across the ER membrane occurs.
5. The signal-cleaving enzyme cuts off the signal peptide.
6. The rest of the completed polypeptide leaves the ribosome and folds.

Targeting Polypeptides to Specific Locations



Protein Folding and Post-translational Modifications

During and after synthesis, a polypeptide chain spontaneously coils and folds into its three-dimensional shape.

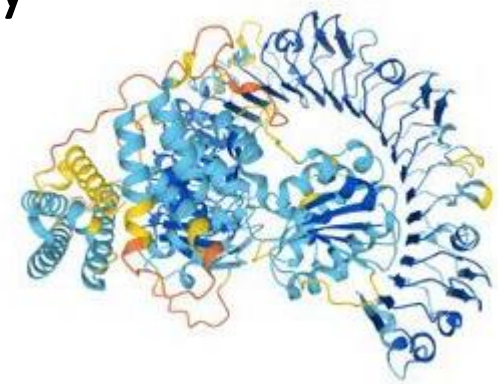
Proteins may also require post-translational modifications before doing their job.

– These modifications may require additions such as sugars, lipids, or phosphate groups to amino acids.

– Enzymes may remove some amino acids from the amino end of the polypeptide chain.

– In some cases, a single polypeptide chain may be enzymatically cleaved into two or more pieces.

– In other cases, two or more polypeptides may join to form a protein with quaternary structure.



Mutations of one or a few nucleotides can affect protein structure and function

Mutations are changes in the genetic material of a cell or virus.

Mutations can be either large-scale, in which long segments of DNA are affected (for example, translocations, duplications, and inversions), or point mutations.

Point mutations are chemical changes in just one base pair of a gene.

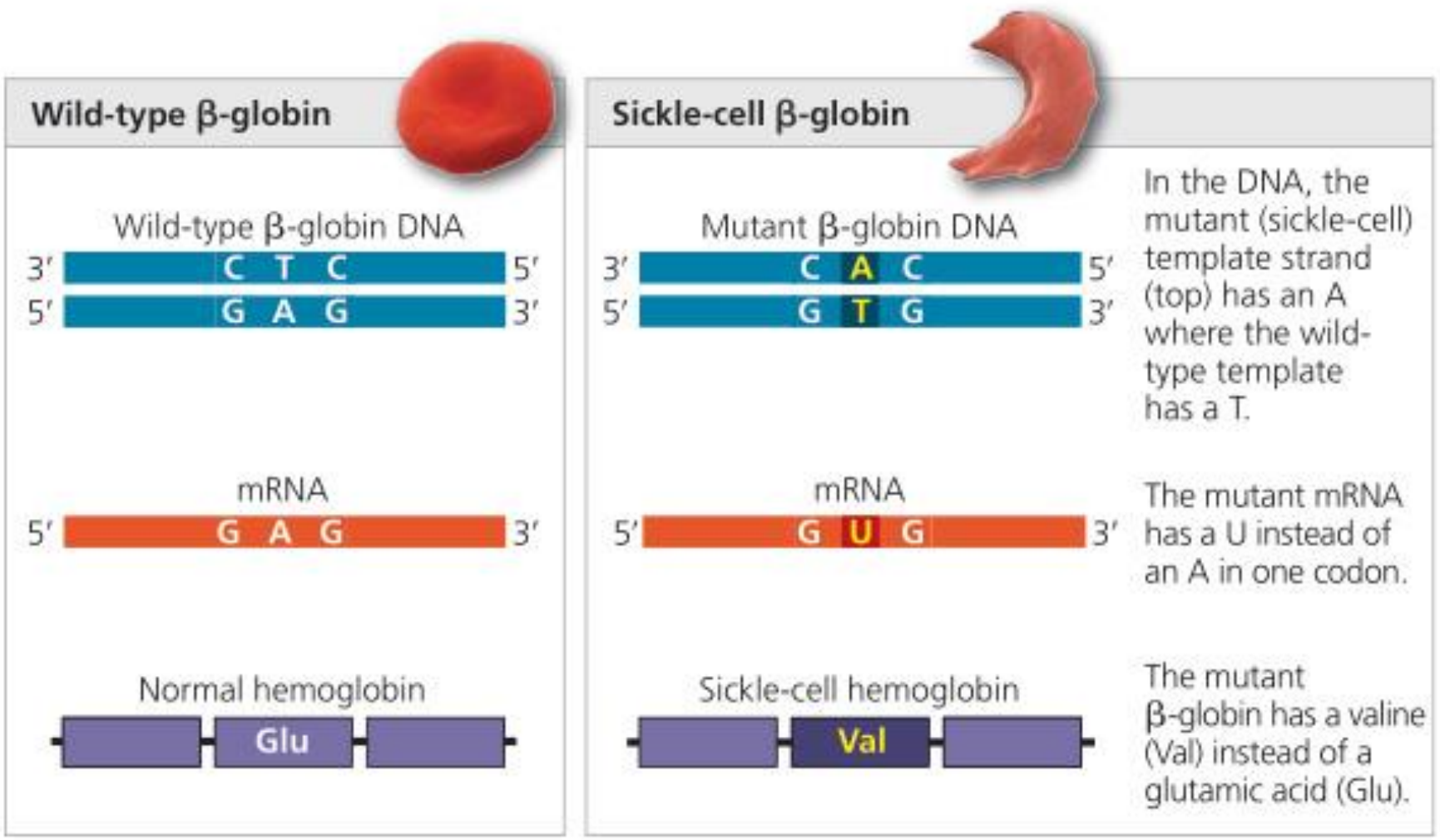
The change of a single nucleotide in a DNA template strand can lead to the production of an abnormal protein.

If a point mutation occurs in a gamete or in a cell that produces gametes, it may be transmitted to future generations.

If the mutation has an adverse effect on the phenotype of a person, the mutant condition is referred to as a **genetic disorder or hereditary disease**.

Mutations of one or a few nucleotides can affect protein structure and function

▼ Figure 17.26 The molecular basis of sickle-cell disease: a point mutation. The allele that causes sickle-cell disease differs from the wild-type (normal) allele by a single DNA nucleotide pair. The micrographs are SEMs of a normal red blood cell (on the left) and a sickled red blood cell (right) from individuals homozygous for wild-type and mutant alleles, respectively.



Types of Point Mutations

Point mutations within a gene can be divided into two general categories

- Base-pair substitutions
- Base-pair insertions or deletions

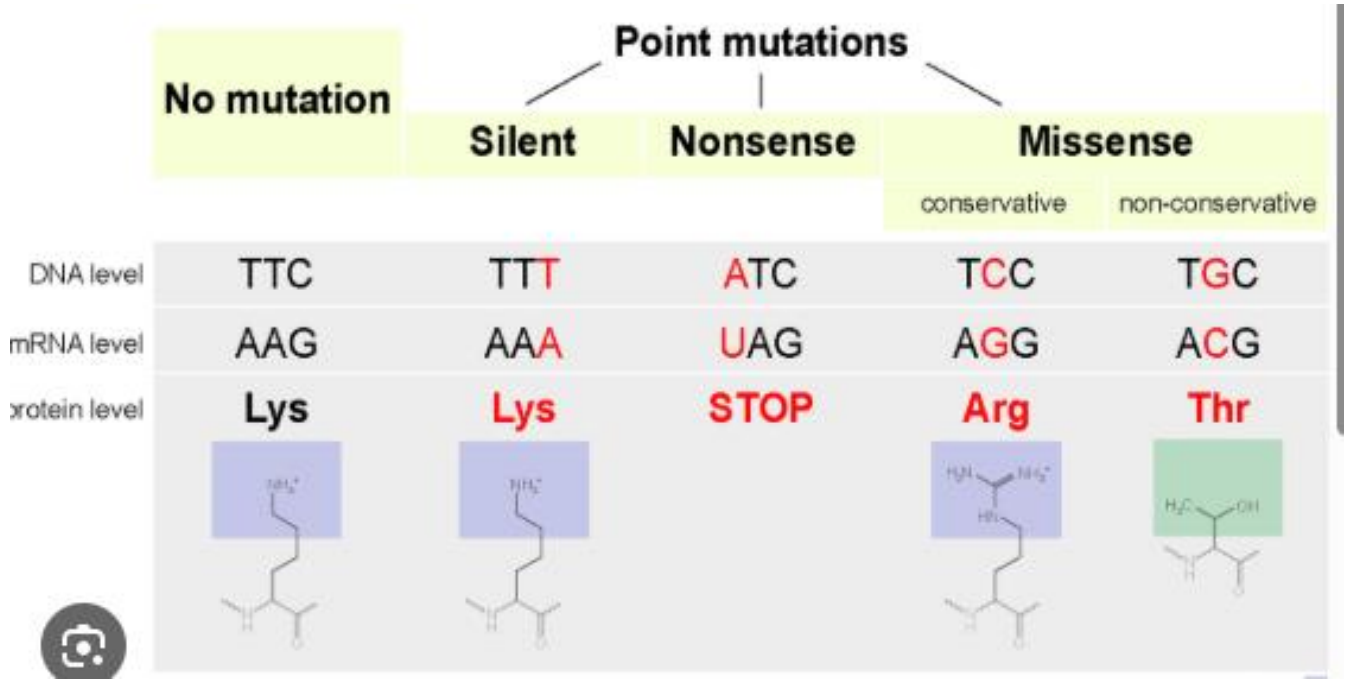


Substitutions

A nucleotide-pair substitution is the replacement of one nucleotide and its partner with another pair of nucleotides.

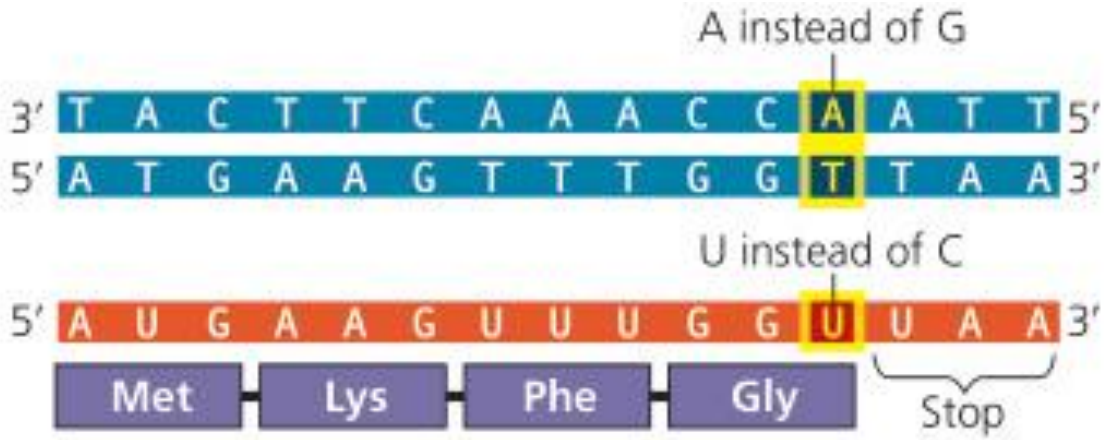
There are three types of substitutions:

- Silent Mutations
- Missense Mutations
- Nonsense Mutations



Silent mutations

Silent mutations occur when a change in a nucleotide pair may transform one codon into another that is translated into the same amino acid.



Silent: No effect on the amino acid sequence

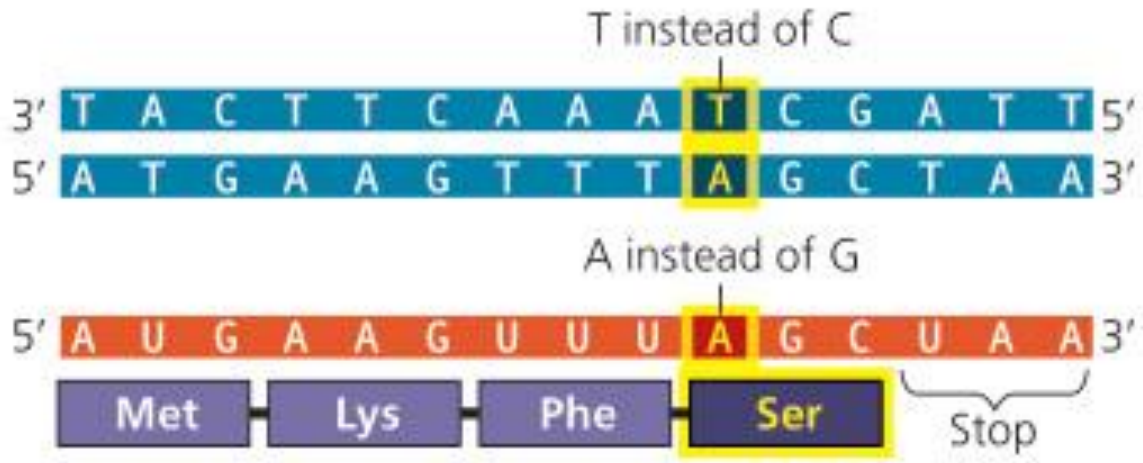
Silent mutations have no observable effect on the phenotype. (Silent mutations can occur outside genes as well.) Interestingly, there is evidence that some silent mutations may indirectly affect where or at what level the gene gets expressed, even though the actual protein is the same.

Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
GCA	AGA	GAC	AAC	UGC	GAA	CAA	GGA	CAC	AUA	UUA	AAA	AUG	UUC	CCA	AGC	ACA	UGG	UAC	GUA	UAA
GCC	AGG	GAU	AAU	UGU	GAG	CAG	GGC	CAU	AUC	UUG	AAG		UUU	CCC	AGU	ACC		UAU	GUC	UAG
GCG	CGA						GGG		AUU	CUA				CCG	UCA	ACG			GUG	UGA
GCU	CGC						GGU			CUC				CCU	UCC	ACU			GUU	
	CGG									CUG					UCG					
	CGU									CUU					UCU					

Missense mutations

Missense mutations occur when substitutions change one amino acid to another. e.g. Sickle-cell disease.

If the new amino acid has properties similar to those of the amino acid it replaces, or it is in a region of the protein where the exact sequence of amino acids is not essential to the protein's function, missense mutations may have little effect on the protein.

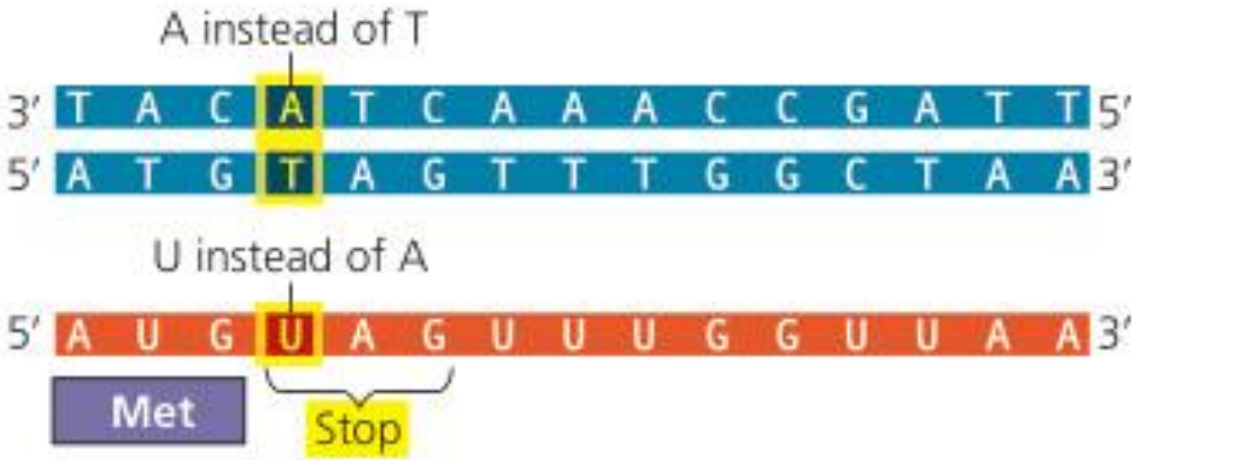


Missense: A range of effects depending on the location within the protein and the identity of the new amino acid

Occasionally, such a mutation leads to an improved protein or one with novel capabilities, but much more often such mutations are neutral or detrimental, leading to a useless or less active protein that impairs cellular function.

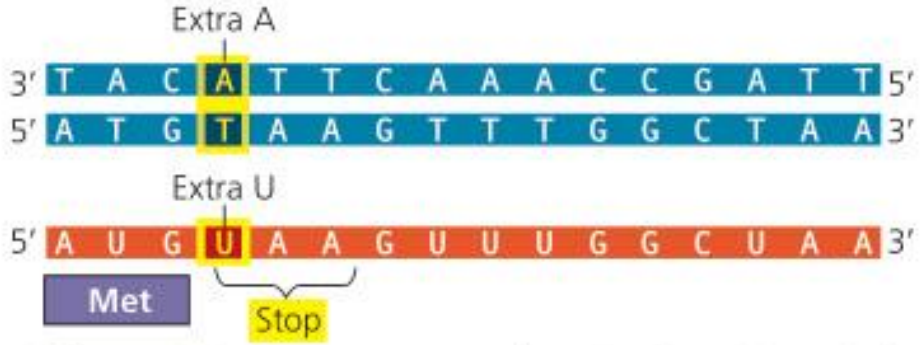
Nonsense mutations

Nonsense mutations occur when substitutions change one amino acid into a stop codon.
This causes translation to be terminated prematurely resulting in a polypeptide shorter than usual. Most nonsense mutations lead to nonfunctional proteins.

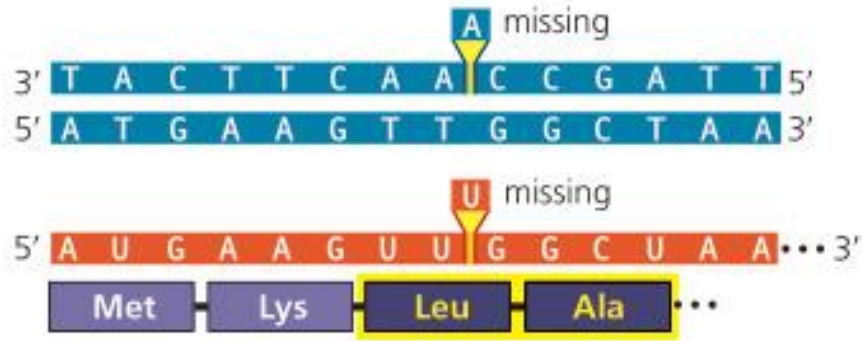


Nonsense: Effect depends on how close the new stop codon is to the beginning of the coding sequence

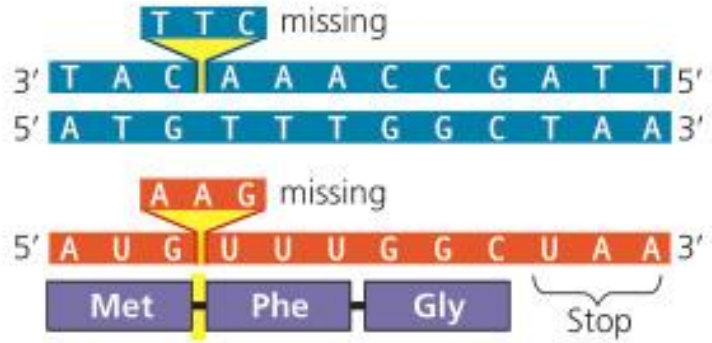
Insertions and Deletions



Frameshift causing immediate nonsense (1-nucleotide-pair insertion)



Frameshift causing extensive missense (1-nucleotide-pair deletion)



3-nucleotide-pair deletion: No frameshift, but one amino acid is missing. A 3-nucleotide-pair insertion (not shown) would lead to an extra amino acid.

Insertions and Deletions

Insertions and deletions are additions or losses of nucleotide pairs in a gene.

These mutations have a disastrous effect on the resulting protein more often than substitutions do.

Insertion or deletion of nucleotides may alter the reading frame, producing a **frameshift mutation** (Unless insertion or deletion mutations occur in multiples of 3, they cause a frameshift mutation). All nucleotides downstream of the deletion or insertion will be improperly grouped into codons.

The result will be extensive missense, ending sooner or later in nonsense—premature termination.

New Mutations and Mutagens

Spontaneous mutations can occur during DNA replication, DNA repair, or DNA recombination.

Mutagens are chemical or physical agents that interact with DNA to cause mutations.

Physical agents include high-energy radiation like X-rays and ultraviolet light.

– Ultraviolet light can cause disruptive thymine dimers in DNA.

• Chemical mutagens cause mutations in different ways.

– Nucleotide analogs are chemicals similar to normal DNA nucleotides and may be substituted into DNA, but they pair incorrectly during DNA replication.

– Other mutagens interfere with DNA replication by inserting themselves into DNA and distorting the double helix.

– Still others cause chemical changes in nucleotide bases that change their pairing properties.

What Is a Gene? Revisiting the Question

The idea of the gene itself is a unifying concept of life

We have considered a gene as:

1. A discrete unit of inheritance
2. A region of specific nucleotide sequence in a chromosome
3. A gene is a region of DNA that can be expressed to produce a final functional product that is either a polypeptide or an RNA molecule.

- All these definitions are useful, depending on the context in which genes are being studied.
- When considering phenotypes, it is often useful to start by focusing on genes that code for polypeptides.
- Proteins, bring about an organism's observable phenotype.

Regulation of Gene Expression

A given type of cell expresses only a subset of its genes.
This is an essential feature in multicellular organisms:
You'd be in trouble if the lens cells in your eyes started expressing the genes for hair proteins, which are normally expressed only in hair follicle cells!
Therefore, gene expression is precisely regulated.

Regulation of Gene Expression

Prokaryotes and eukaryotes alter gene expression in response to their changing environment.

In multicellular eukaryotes, gene expression regulates development and is responsible for differences in cell types.

RNA molecules play many roles in regulating gene expression in eukaryotes.

Bacteria often respond to environmental change by regulating transcription

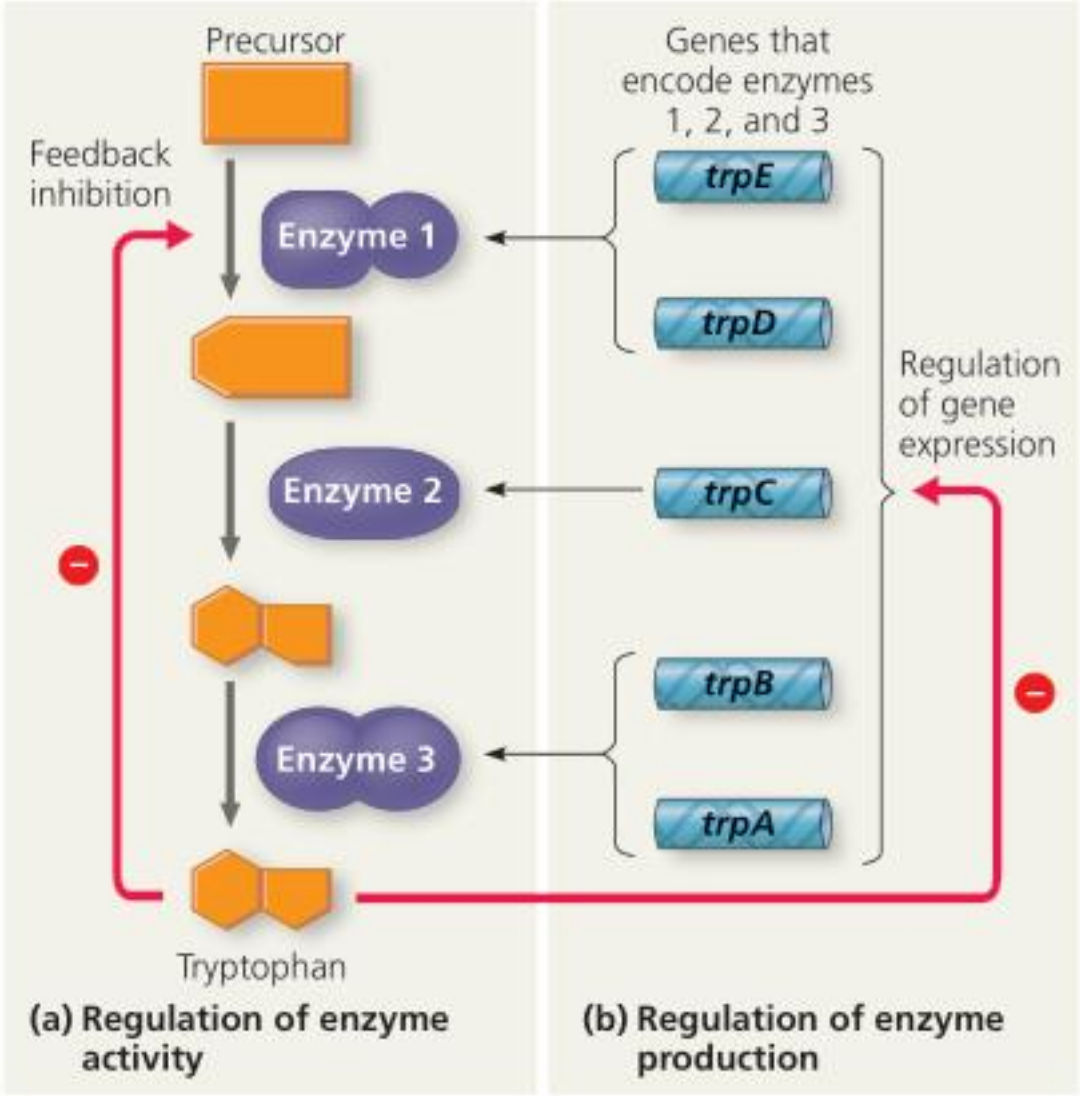
Bacterial cells that can conserve resources and energy have a selective advantage over cells that are unable to do so.
Thus, natural selection has favored bacteria that produce only the products needed by that cell.

- A cell can regulate the production of enzymes by feedback inhibition or by gene regulation.

- Gene expression in bacteria is controlled by the operon model.

Regulation of Gene Expression in bacteria

▼ Figure 18.2 Regulation of a metabolic pathway. In the pathway for tryptophan synthesis, an abundance of tryptophan can both **(a)** inhibit the activity of the first enzyme in the pathway (feedback inhibition), a rapid response, and **(b)** repress expression of the genes encoding all subunits of the enzymes in the pathway, a longer-term response. The symbol ⊖ stands for inhibition.



First, cells can adjust the activity of enzymes already present. The activity of the first enzyme in the pathway is inhibited by the pathway's end product—tryptophan, in this case. (Figure 18.2a). Thus, if tryptophan accumulates in a cell, it shuts down the synthesis of more tryptophan by inhibiting enzyme activity. Such feedback inhibition, typical of anabolic (biosynthetic) pathways, allows a cell to adapt to short-term fluctuations in the supply of a substance it needs.

Second, cells can adjust the production level of certain enzymes via a genetic mechanism; that is, they can regulate the expression of the genes encoding the enzymes. If, in our example, the environment provides all the tryptophan the cell needs, the cell stops making the enzymes that catalyze the synthesis of tryptophan (Figure 18.2b). In this case, the control of enzyme production occurs at the level of transcription, the synthesis of messenger RNA from the genes that code for these enzymes.

Operons: The Basic Concept

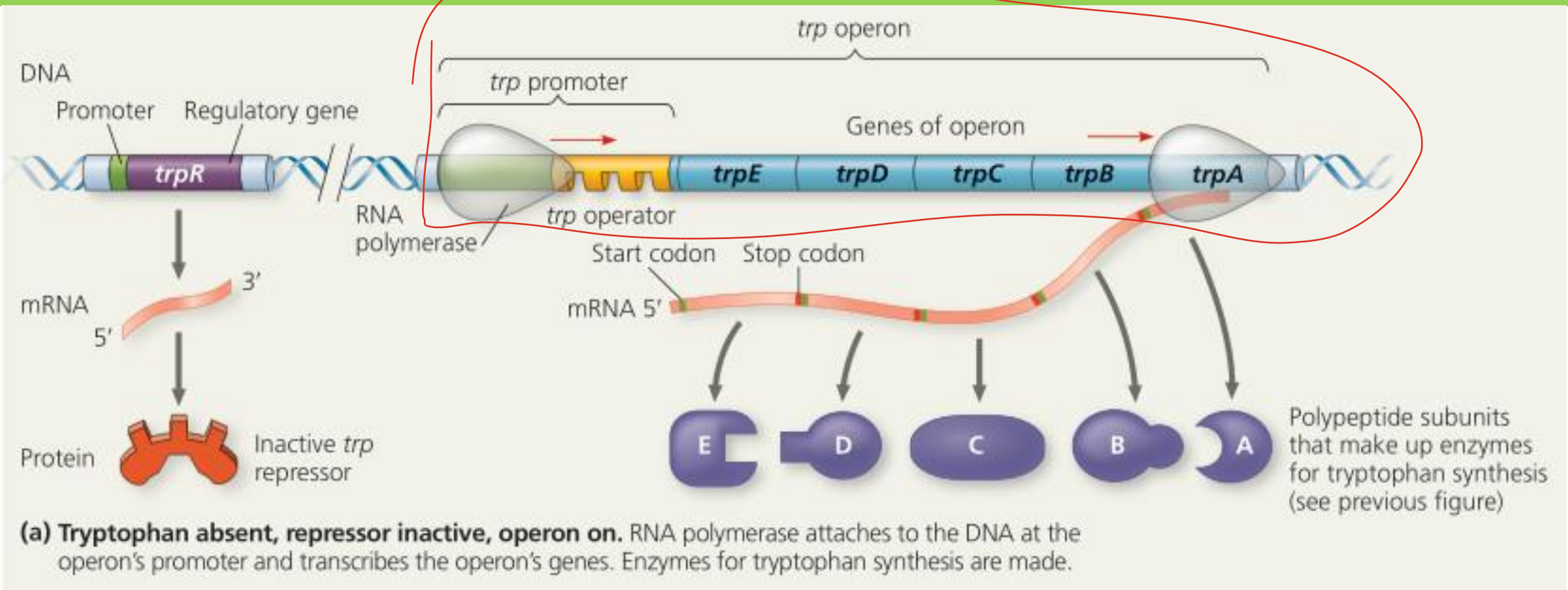
A cluster of functionally related genes can be under coordinated control by a single on-off “switch”.

The regulatory “switch” is a segment of DNA called an **operator** usually positioned within the promoter.

An operon is the entire stretch of DNA that includes the operator, the promoter, and the genes that they control.

For example the five genes that function in the production of tryptophan form a cluster on bacterial genome and they are under the control of the same operator.

Operons: The Basic Concept



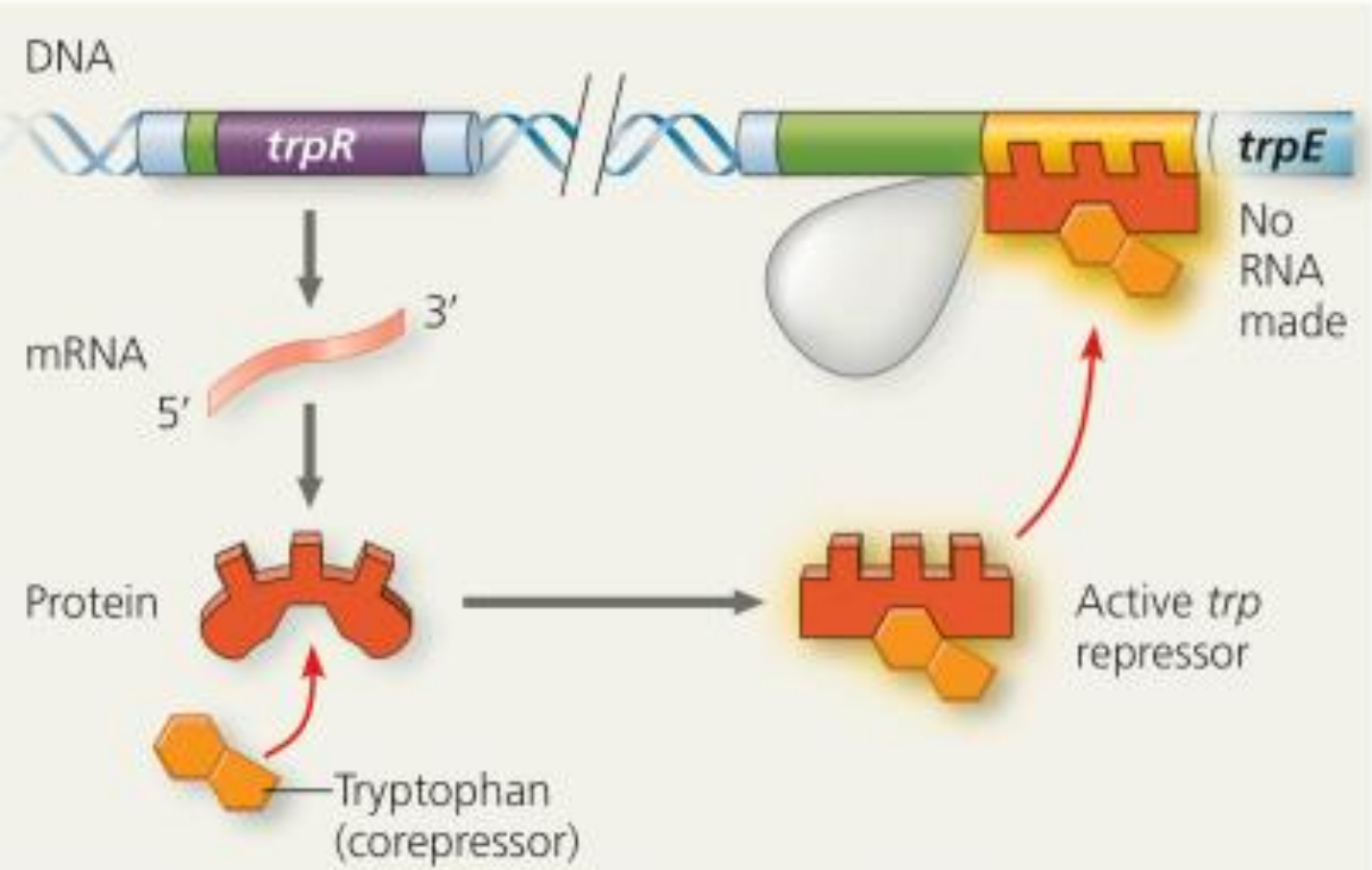
(a) Tryptophan absent, repressor inactive, operon on. RNA polymerase attaches to the DNA at the operon's promoter and transcribes the operon's genes. Enzymes for tryptophan synthesis are made.

The operon can be switched off by a protein repressor

The repressor prevents gene transcription by binding to the operator and blocking RNA polymerase

The repressor is the product of a separate regulatory gene

Operons: The Basic Concept



(b) Tryptophan present, repressor active, operon off. As tryptophan accumulates, it inhibits its own production by activating the repressor protein, which binds to the operator, blocking transcription. Enzymes for tryptophan synthesis are not made.

The operon can be switched off by a protein repressor

The repressor prevents gene transcription by binding to the operator and blocking RNA polymerase

The repressor is the product of a separate regulatory gene

The repressor can be in an active or inactive form, depending on the presence of other molecules.

A **corepressor** is a molecule that cooperates with a repressor protein to switch an operon off.

Operons: The Basic Concept

By default the trp operon is on and the genes for tryptophan synthesis are transcribed.

When tryptophan is present, it binds to the trp repressor protein, which binds to the **operator** and turns the operon off .

Positioned within the promoter or, in some cases, between the promoter and the enzyme-coding genes, the operator controls the access of RNA polymerase to the genes.

A repressor protein is specific for the operator of a particular operon.
The repressor is active only in the presence of its corepressor tryptophan; thus the trp operon is turned off (repressed) if tryptophan levels are high.

Operons: The Basic Concept

Why is the trp operon not switched off permanently?

First, the binding of repressors to operators is reversible. An operator alternates between two states:
one with the repressor bound and one without the repressor bound.
The relative duration of the repressor-bound state is higher when more active repressor molecules are present.
Second, the trp repressor, like most regulatory proteins, is
an allosteric protein, with two alternative shapes: active and
İnactive. The trp repressor is synthesized in the inactive form, which has little affinity for the trp operator. Only when a tryptophan molecule binds to the trp repressor at an allosteric site does the repressor protein change to the active form that can attach to the operator, turning the operon off.

Operons: The Basic Concept

Tryptophan functions in this system as a **corepressor**, a small molecule that cooperates with a repressor protein to switch an operon off.
As tryptophan accumulates, more tryptophan molecules associate with trp repressor molecules, which can then bind to the trp operator and shut down production of the tryptophan pathway enzymes.
If the cell's tryptophan level drops, many fewer trp repressor proteins would have tryptophan bound, rendering them inactive; they would dissociate from the operator, allowing transcription of the operon's genes to resume.

The trp operon is one example of how gene expression can respond to changes in the cell's internal and external environment (homeostasis).

Repressible and Inducible Operons: Two Types of Negative Gene Regulation

A **repressible operon** is one that is usually on; binding of a repressor to the operator shuts off transcription.
The trp operon is a repressible operon.

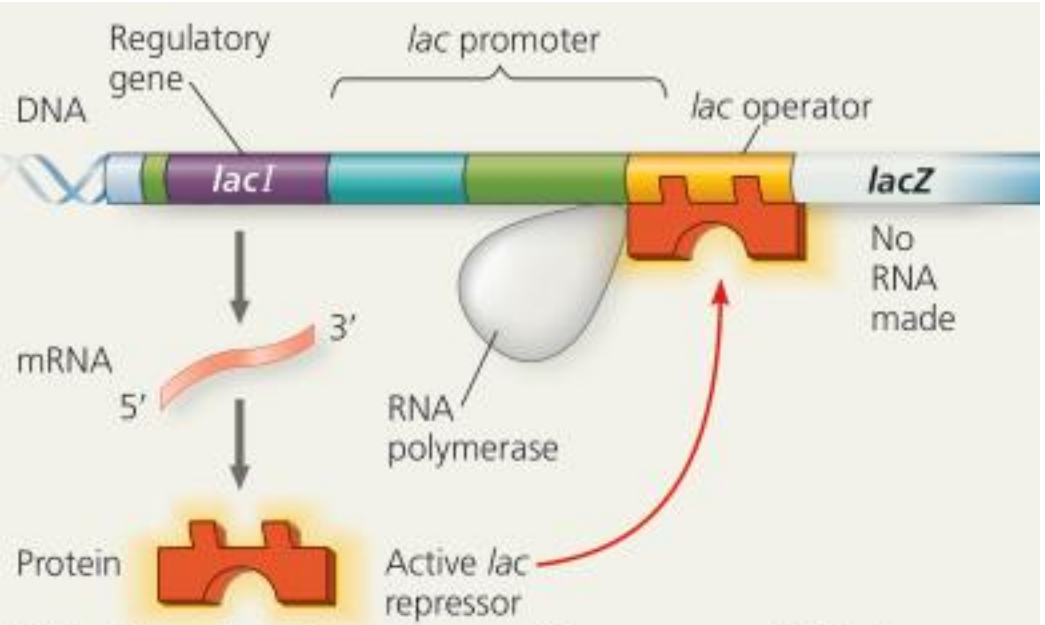
An **inducible operon** is one that is usually off; a molecule called an inducer inactivates the repressor and turns on transcription.
The lac operon is an inducable operon.

Inducible Operons

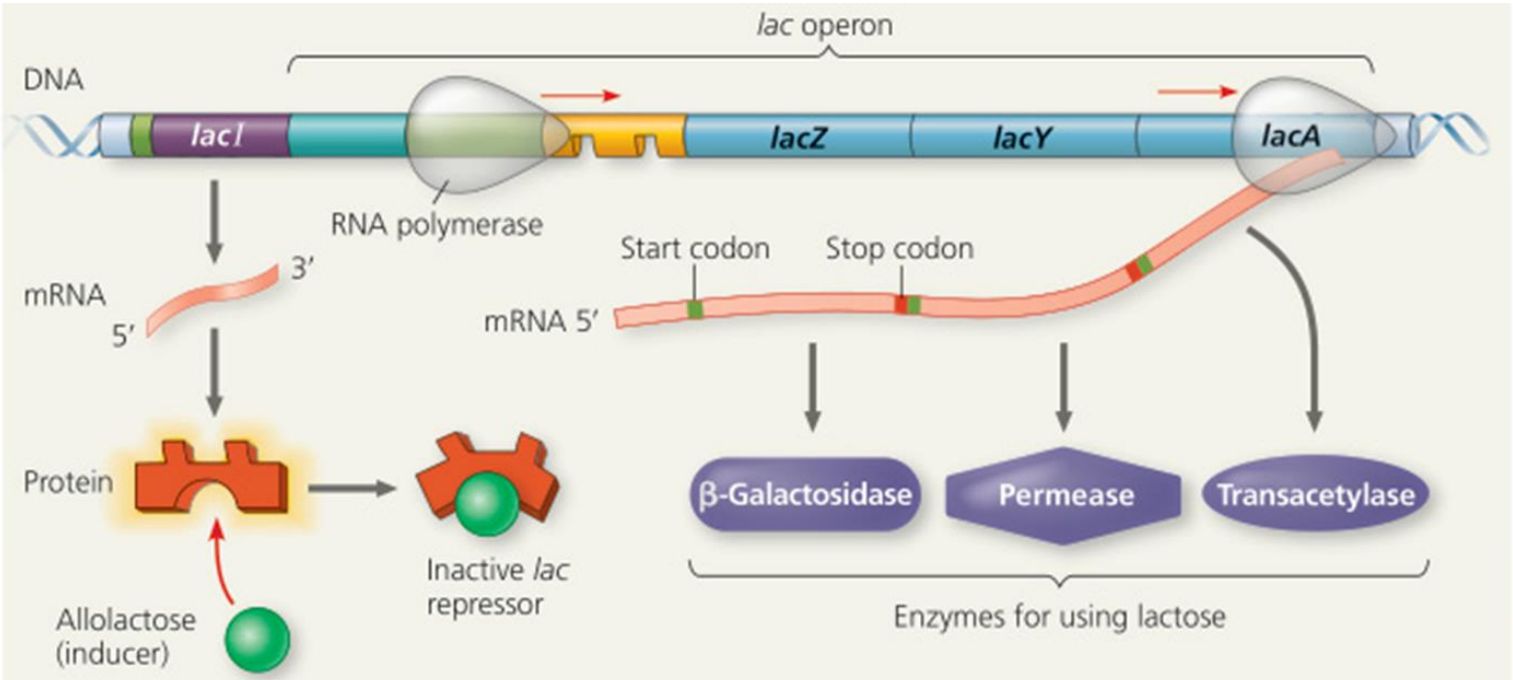
The lac operon contains genes that code for enzymes used in the hydrolysis and metabolism of lactose.

By itself, the lac repressor is active and switches the lac operon off.

A molecule called an inducer inactivates the repressor to turn the lac operon on.



(a) Lactose absent, repressor active, operon off. The *lac* repressor is innately active, and in the absence of lactose it switches off the operon by binding to the operator. The enzymes for using lactose are not made.



(b) Lactose present, repressor inactive, operon on. Allolactose, an isomer of lactose, binds to the repressor, inactivating it and "derepressing" the operon. The inactive repressor cannot bind to the operator, and so the genes of the *lac* operon are transcribed, and the enzymes for using lactose are made.

Repressible and Inducible Operons: Two Types of Negative Gene Regulation

Inducible enzymes usually function in catabolic pathways; their synthesis is induced by a chemical signal.

Repressible enzymes usually function in anabolic pathways; their synthesis is repressed by high levels of the end product.

Regulation of the trp and lac operons involves negative control of genes because operons are switched off by the active form of the repressor.

Positive Gene Regulation

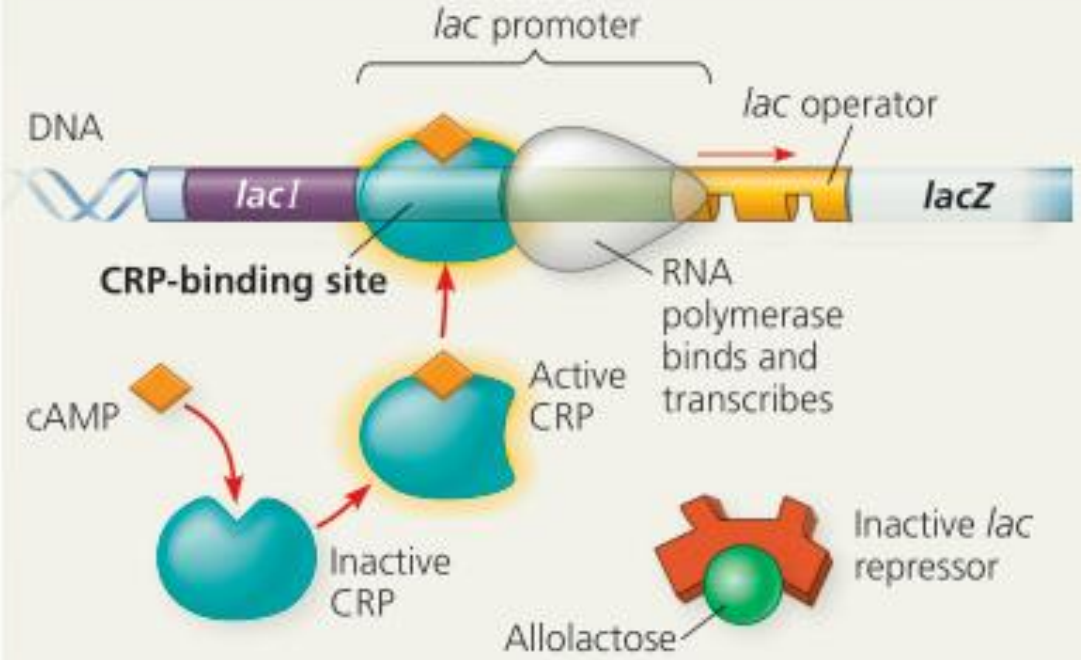
Gene regulation is said to be positive only when a regulatory protein interacts directly with the genome to increase transcription.

Some operons are subject to positive control through a stimulatory protein.
E.g. cAMP receptor protein (CRP), is an activator, a protein that binds to DNA and stimulates transcription of a gene.

When glucose (a preferred food source of E. coli) is scarce cyclic AMP accumulates and CRP is activated by binding with cyclic AMP (cAMP).

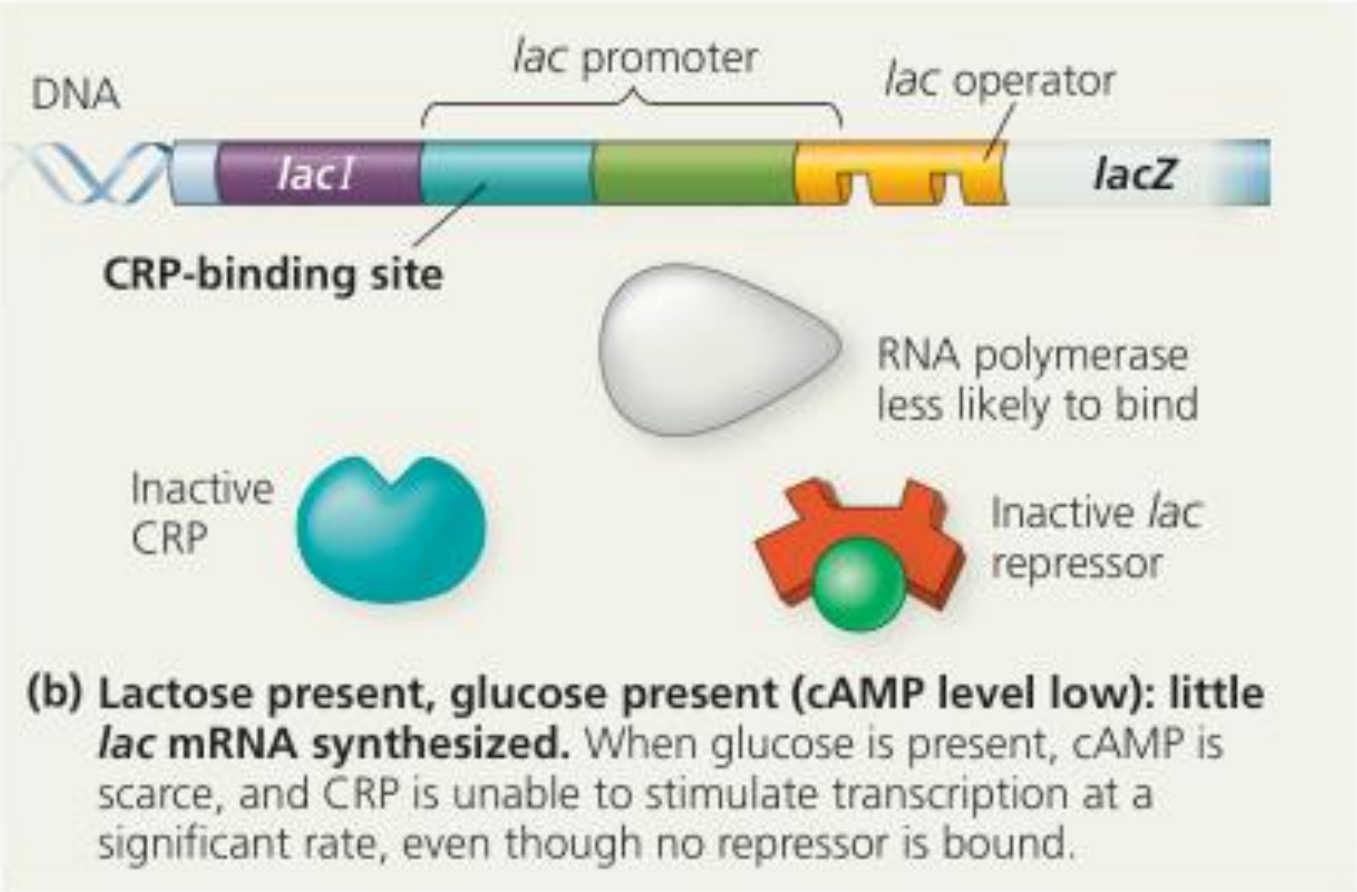
Activated CRP attaches to the promoter of the lac operon and increases the affinity of RNA polymerase, thus accelerating transcription.

▼ Figure 18.5 Positive control of the lac operon by cAMP receptor protein (CRP). RNA polymerase has high affinity for the *lac* promoter only when CRP is bound to a DNA site at the upstream end of the promoter. CRP, in turn, attaches to its DNA site only when associated with cyclic AMP (cAMP), whose concentration in the cell rises when the glucose concentration falls. Thus, when glucose is present, even if lactose is also available, the cell preferentially catabolizes glucose and makes very low levels of the enzymes for using lactose.



(a) Lactose present, glucose scarce (cAMP level high): abundant *lac* mRNA synthesized. If glucose is scarce, the high level of cAMP activates CRP, which binds to the promoter and increases RNA polymerase binding there. The *lac* operon produces large amounts of mRNA coding for the enzymes that the cell needs for use of lactose.

Positive Gene Regulation



(b) Lactose present, glucose present (cAMP level low): little *lac* mRNA synthesized. When glucose is present, cAMP is scarce, and CRP is unable to stimulate transcription at a significant rate, even though no repressor is bound.

When glucose levels increase, CRP detaches from the lac operon, and transcription returns to a normal rate

CAP helps regulate other operons that encode enzymes used in catabolic pathways

The lac operon is under dual control: negative control by the lac repressor and positive control by CRP.

Whether or not transcription occurs is controlled by allolactose:

Without allolactose, the lac repressor is active and the operon is off (transcription does not occur); with allolactose, the lac repressor is inactive and the operon is on (transcription occurs). The rate of transcription is controlled by whether CRP has cAMP bound to it: With bound cAMP, the rate is high; without it, the rate is low. It is as though the operon has both an on-off switch and a volume control.

Eukaryotic gene expression is regulated at many stages

All organisms must regulate which genes are expressed at any given time. Both unicellular organisms and the cells of multicellular organisms continually turn genes on and off in response to signals from their external and internal environments.

Regulation of gene expression is also essential for cell specialization in multicellular organisms, which are made up of different types of cells.

To perform its own distinct role, each cell type must maintain a specific program of gene expression in which certain genes are expressed and others are not.